MUTUAL FUND ANALYTICS PLATFORM

End-to-End Analytics and Business Intelligence Solution for the Indian Mutual Fund Industry

Bluestock Fintech Capstone Project

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Prepared For: Bluestock Fintech

Tools & Technologies Used

Python • Pandas • NumPy • SQLite • SQL • Power BI • Git • GitHub

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Project Scope

- **Data Ingestion**
- **Data Cleaning & Transformation**
- **Exploratory Data Analysis (EDA)**
- **Performance Analytics**
- **Advanced Analytics**
- **Database Design & SQL Analytics**
- **Interactive Power BI Dashboard**
- **Business Intelligence Reporting**

Project Duration

May 2026 – June 2026

Final Project Report

June 2026

1. Executive Summary

1.1 Project Background

The mutual fund industry generates large volumes of data related to fund performance, investor transactions, SIP contributions, benchmark indices, and Assets Under Management (AUM). Analyzing these datasets manually can be challenging due to their volume, complexity, and continuous growth. This project was developed to address these challenges by creating a centralized analytics platform capable of transforming raw financial data into meaningful business insights.

1.2 Project Objectives

The primary objective of this project is to build an end-to-end Mutual Fund Analytics Platform that integrates data engineering, analytics, database management, and business intelligence techniques. The platform aims to analyze industry growth, evaluate fund performance, understand investor behavior, monitor SIP trends, and provide actionable insights through interactive dashboards.

1.3 Methodology Overview

The project follows a structured analytics workflow consisting of data ingestion, data cleaning, transformation, exploratory data analysis, database creation, advanced analytics, and dashboard development. Multiple datasets including fund master data, NAV history, performance metrics, SIP statistics, benchmark indices, category inflows, and investor transactions were processed using Python and stored in a SQLite database for analysis.

Exploratory Data Analysis (EDA) was conducted to identify patterns and trends within the datasets. Performance analytics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to assess mutual fund performance. Interactive visualizations were then developed in Power BI to present insights in a user-friendly format.

1.4 Key Outcomes

The project successfully delivered a comprehensive analytics platform capable of monitoring mutual fund industry performance and investor activity. The final solution includes cleaned datasets, a structured database, analytical notebooks, SQL-based insights, and a multi-page Power BI dashboard.

The dashboard enables users to analyze industry AUM growth, SIP inflows, fund performance metrics, investor transaction trends, category-level inflows, and benchmark comparisons. The project demonstrates the practical application of modern analytics tools and business intelligence techniques in solving real-world financial analysis problems.

2. Project Overview

2.1 Introduction

The Mutual Fund Analytics Platform is an end-to-end data analytics and business intelligence project developed to analyze the Indian mutual fund industry. The project integrates multiple datasets related to mutual funds, investor transactions, market benchmarks, SIP inflows, Assets Under Management (AUM), and fund performance metrics. The primary objective is to convert raw financial data into meaningful insights through data processing, analytics, and interactive visualization techniques.

The platform enables users to understand industry growth, evaluate fund performance, monitor investor behavior, and analyze market trends using a centralized analytical framework.

2.2 Project Scope

The scope of the project covers the complete data analytics lifecycle, starting from data ingestion and ending with interactive dashboard development. The project includes data cleaning, transformation, exploratory data analysis, database creation, SQL analytics, performance evaluation, advanced analytics, and business intelligence reporting.

The analysis focuses on key areas such as fund performance measurement, SIP growth analysis, benchmark comparison, category-wise inflows, investor demographics, transaction patterns, and overall industry trends.

2.3 Project Workflow

The project was executed through the following stages:

1. Data Collection and Ingestion
2. Data Cleaning and Transformation
3. Exploratory Data Analysis (EDA)
4. Database Design and SQL Analytics
5. Performance Analytics
6. Advanced Analytics
7. Power BI Dashboard Development
8. Business Insight Generation
9. Reporting and Documentation

Each stage was designed to ensure data quality, analytical accuracy, and effective visualization of results.

2.4 Expected Outcomes

The expected outcomes of the project include:

- Creation of a structured mutual fund analytics database.
- Development of reusable ETL pipelines.
- Identification of industry growth trends and patterns.
- Evaluation of fund performance using risk and return metrics.
- Analysis of investor transaction behavior and SIP participation.
- Development of an interactive Power BI dashboard.
- Generation of actionable insights for decision-making.

The final solution provides a comprehensive analytical view of the mutual fund ecosystem and demonstrates the practical application of data analytics and business intelligence technologies.

3. Problem Statement

3.1 Industry Background

The Indian mutual fund industry has experienced significant growth over the past decade, driven by increasing investor awareness, digital investment platforms, and the rising popularity of Systematic Investment Plans (SIPs). As the number of mutual fund schemes, investors, and transactions continues to grow, large volumes of data are generated across various segments of the industry.

This data includes fund performance metrics, Net Asset Value (NAV) history, Assets Under Management (AUM), investor transactions, SIP contributions, benchmark index performance, category-wise inflows, and folio statistics. Managing and analyzing such diverse datasets is essential for understanding industry trends and making informed investment decisions.

3.2 Existing Challenges

Despite the availability of large amounts of mutual fund data, several challenges exist in converting this data into meaningful insights.

- Data is often distributed across multiple sources and formats.
- Manual analysis of large datasets is time-consuming and inefficient.
- Identifying performance trends across multiple funds can be difficult.
- Comparing fund returns with benchmark indices requires extensive calculations.

- Investor transaction patterns and SIP trends are not easily visible without structured analysis.
- Decision-makers often lack a centralized platform for monitoring industry performance.

These challenges limit the ability of analysts and investors to make timely, data-driven decisions.

3.3 Problem Identification

The absence of an integrated analytics platform makes it difficult to analyze mutual fund industry performance from a single viewpoint. Raw datasets alone do not provide sufficient business value without proper cleaning, transformation, analysis, and visualization.

There is a need for a solution that can consolidate multiple mutual fund datasets, perform advanced analytical calculations, and present insights through intuitive visualizations. Such a platform should support performance evaluation, investor analysis, industry monitoring, and trend identification while reducing the complexity of manual analysis.

3.4 Proposed Solution

To address these challenges, the Mutual Fund Analytics Platform was developed as an end-to-end analytics and business intelligence solution.

The proposed solution integrates data engineering, database management, analytics, and visualization techniques to create a centralized platform for mutual fund analysis. The system performs data ingestion, cleaning, transformation, exploratory analysis, performance evaluation, and advanced analytics before presenting the results through an interactive Power BI dashboard.

The platform enables users to monitor industry growth, evaluate fund performance using risk and return metrics, analyze investor behavior, track SIP trends, compare benchmark performance, and generate actionable insights for decision-making. By providing a unified analytical environment, the solution improves efficiency, enhances data visibility, and supports informed investment analysis.

4. Project Objectives

4.1 Primary Objectives

The primary objective of this project is to develop a comprehensive Mutual Fund Analytics Platform that enables efficient analysis of the Indian mutual fund industry through data analytics and business intelligence techniques.

The project aims to consolidate multiple mutual fund datasets into a centralized analytical environment and transform raw data into meaningful insights that support informed decision-making.

The key primary objectives include:

- To build an end-to-end data analytics solution for mutual fund analysis.
- To integrate multiple datasets related to funds, investors, SIPs, benchmarks, and industry statistics.
- To evaluate mutual fund performance using risk and return metrics.
- To develop interactive dashboards for data visualization and reporting.
- To generate actionable insights from financial and investor data.

4.2 Secondary Objectives

In addition to the primary goals, the project focuses on several supporting objectives that enhance analytical capabilities and data accessibility.

These objectives include:

- To design and implement an efficient ETL pipeline.
- To clean, validate, and standardize raw datasets.
- To create a structured SQLite database for analytical processing.
- To perform exploratory data analysis on industry datasets.
- To identify trends in AUM growth, SIP inflows, and investor participation.
- To analyze benchmark performance and market movements.
- To evaluate category-wise fund inflows and performance patterns.
- To study investor transaction behavior across demographics and regions.
- To implement drill-through and interactive dashboard functionality in Power BI.

4.3 Expected Benefits

The successful implementation of this project is expected to provide several benefits for users and stakeholders.

These benefits include:

- Improved visibility into mutual fund industry trends.
- Faster analysis of fund performance and investment metrics.
- Better understanding of investor behavior and transaction patterns.
- Centralized access to key financial indicators and performance measures.
- Enhanced reporting through interactive dashboards and visualizations.

- Support for data-driven investment analysis and decision-making.
- Practical demonstration of data engineering, analytics, SQL, and business intelligence concepts.

The final outcome is a scalable analytics platform capable of transforming complex mutual fund data into meaningful business insights through an intuitive and interactive reporting environment.

5. Dataset Description

5.1 Overview of Datasets

The Mutual Fund Analytics Platform utilizes multiple datasets covering different aspects of the Indian mutual fund ecosystem. These datasets provide information related to fund characteristics, performance metrics, investor transactions, SIP activities, benchmark indices, industry growth, and category-wise fund flows.

The integration of these datasets enables comprehensive analysis of mutual fund performance, investor behavior, market trends, and industry growth patterns.

The project dataset repository consists of the following major datasets:

- Fund Master Dataset
- NAV History Dataset
- Performance Dataset
- Investor Transactions Dataset
- SIP Dataset
- AUM Dataset
- Category Dataset
- Benchmark Dataset
- Folio Dataset

Each dataset contributes unique information required for analytical processing and dashboard development.

Dataset Name	Records	Key Attributes	Purpose
Fund Master	40	amfi_code, scheme_name, category	Fund Information
NAV History	XXXX	date, nav	NAV Trend Analysis
Performance	XXXX	CAGR, Sharpe, Sortino	Performance Analytics
Transactions	XXXX	transaction_type, amount_inr	Investor Analytics
SIP	XXXX	sip_inflow_crore	SIP Analysis
AUM	XXXX	aum_lakh_crore	Industry Growth Analysis
Category	XXXX	category, net_inflow_crore	Category Analysis
Benchmark	XXXX	index_name, close_value	Market Comparison
Folio	XXXX	folios_crore	Investor Participation

5.2 Fund Master Dataset

The Fund Master Dataset serves as the primary reference dataset for mutual fund schemes.

This dataset contains scheme-level information including:

- AMFI Code
- Scheme Name
- Fund House
- Category
- Sub-Category
- Plan Type
- Launch Date
- Benchmark Information
- Expense Ratio
- Exit Load

- Fund Manager Details
- Risk Category

The dataset is used for fund identification, classification, filtering, and dashboard navigation.

```
Figure 5.1: Fund Master Dataset Overview
========================================================================================================================
Dataset Shape: (40, 15)
========================================================================================================================
 amfi_code       fund_house                                        scheme_name category   sub_category    plan launch_date
                benchmark  expense_ratio_pct  exit_load_pct  min_sip_amount  min_lumpsum_amount    fund_manager risk_category
sebi_category_code
    119551  SBI Mutual Fund         SBI Bluechip Fund - Regular Plan - Growth   Equity      Large Cap Regular  2006-02-14
        NIFTY 100 TRI              1.54            1.0             500                    1000   Sohini Andani      Moderate
          EC01
    119552  SBI Mutual Fund          SBI Bluechip Fund - Direct Plan - Growth   Equity      Large Cap  Direct  2013-01-01
        NIFTY 100 TRI              0.66            1.0             500                    1000   Sohini Andani      Moderate
          EC01
    119598  SBI Mutual Fund        SBI Small Cap Fund - Regular Plan - Growth   Equity      Small Cap Regular  2009-09-09
     BSE 250 SmallCap TRI          1.43            1.0             500                    1000   R. Srinivasan     Very High
          EC03
    119599  SBI Mutual Fund         SBI Small Cap Fund - Direct Plan - Growth   Equity      Small Cap  Direct  2013-01-01
     BSE 250 SmallCap TRI          0.72            1.0             500                    1000   R. Srinivasan     Very High
          EC03
    119120  SBI Mutual Fund       SBI Magnum Gilt Fund - Regular Plan - Growth    Debt           Gilt Regular  2000-12-30
CRISIL Dynamic Gilt Index          0.77            0.0             500                    1000    Dinesh Ahuja           Low
          DC02
```

5.3 NAV History Dataset

The NAV History Dataset contains historical Net Asset Value records for mutual fund schemes.

Key attributes include:

- Date
- AMFI Code
- NAV Value

This dataset is used to analyze fund performance over time, calculate returns, evaluate drawdowns, and compare schemes against benchmark indices.

Historical NAV data forms the foundation for several performance metrics used throughout the project.

```
Figure 5.2: NAV History Dataset Sample
==================================================
Dataset Shape: (46000, 3)
==================================================
 amfi_code        date       nav
    100016  2022-01-03  520.4608
    100016  2022-01-04  515.0971
    100016  2022-01-05  521.7239
    100016  2022-01-06  515.7880
    100016  2022-01-07  515.1639
    100016  2022-01-10  510.7136
    100016  2022-01-11  513.5542
    100016  2022-01-12  512.3195
    100016  2022-01-13  510.2445
    100016  2022-01-14  514.3636
```

5.4 Performance Dataset

The Performance Dataset contains analytical and risk-adjusted performance measures calculated for each mutual fund scheme.

Key metrics include:

- 1-Year Return
- 3-Year Return
- 5-Year Return
- CAGR
- Standard Deviation
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Risk Grade

This dataset enables comparative analysis of fund performance and risk characteristics.

5.5 Investor Transactions Dataset

The Investor Transactions Dataset contains transaction-level investment activity performed by investors.

The dataset includes:

- Transaction Date
- Transaction Type
- Investment Amount
- State
- City Tier
- Age Group
- Annual Income

- KYC Status

The dataset supports analysis of investor behavior, transaction trends, geographic participation, and demographic patterns.

5.6 SIP Dataset

The SIP Dataset contains Systematic Investment Plan (SIP) statistics and growth indicators.

Key information includes:

- SIP Inflows
- SIP Assets Under Management
- Active SIP Accounts
- Monthly SIP Trends

This dataset is used to analyze investor participation through SIP investments and identify long-term industry growth trends.

5.7 AUM Dataset

The Assets Under Management (AUM) Dataset provides information about the total assets managed by the mutual fund industry.

Key fields include:

- Reporting Period
- Total AUM
- Fund House AUM

The dataset is used to evaluate industry growth and compare the market share of different Asset Management Companies (AMCs).

5.8 Category Dataset

The Category Dataset contains category-level inflow and outflow information across different mutual fund segments.

Examples of categories include:

- Large Cap
- Mid Cap
- Small Cap
- Flexi Cap

- Liquid Funds
- Sectoral/Thematic Funds
- Hybrid Funds

This dataset helps identify investor preferences and category-wise investment trends.

5.9 Benchmark Dataset

The Benchmark Dataset contains market index performance data used for comparison with mutual fund schemes.

The dataset includes:

- Date
- Index Name
- Closing Value

Benchmark indices such as Nifty 50 and Nifty 100 are used to evaluate fund performance against market movements.

5.10 Folio Dataset

The Folio Dataset contains information regarding mutual fund investor accounts.

Key attributes include:

- Reporting Period
- Number of Folios
- Category-Level Folio Counts

This dataset helps measure investor participation and growth in mutual fund adoption over time.

5.11 Data Quality Assessment

Before analytical processing, all datasets underwent a comprehensive data quality assessment process.

The assessment included:

- Missing Value Detection
- Duplicate Record Identification
- Data Type Validation
- Range Validation

- Consistency Checks
- Date Standardization
- Outlier Detection

Data cleaning and validation ensured that the datasets were accurate, reliable, and suitable for downstream analytics and dashboard development.

The final cleaned datasets were stored in a structured SQLite database and used for all subsequent analysis and visualization tasks.

Dataset Name	Missing Values	Duplicate Records	Data Quality Status
Fund Master	0	0	Clean
NAV History	0	0	Clean
Performance	0	0	Clean
Transactions	0	0	Clean
SIP	12	0	Minor Missing Values Handled
AUM	0	0	Clean
Category	0	0	Clean
Benchmark	0	0	Clean
Folio	0	0	Clean

6. Data Architecture

6.1 Architecture Overview

A structured data architecture was designed to efficiently manage, store, and analyze mutual fund datasets. The architecture follows a centralized analytics approach where data from multiple sources is collected, processed, stored, and visualized through a business intelligence platform.

The architecture consists of four major layers:

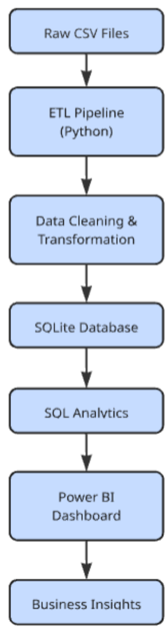
1. Data Source Layer
2. Data Processing Layer
3. Database Layer

4. Visualization Layer

This layered approach improves scalability, maintainability, and analytical efficiency throughout the project lifecycle.

6.2 Data Flow Architecture

The project follows a sequential data flow process from raw data collection to dashboard visualization.



Step 1: Data Collection

Raw datasets were collected from multiple mutual fund-related sources and stored as CSV files.

The collected datasets include:

- Fund Master Data
- NAV History
- Performance Metrics
- Investor Transactions
- SIP Statistics
- AUM Records

- Category Inflows
- Benchmark Data
- Folio Statistics

Step 2: Data Processing

Python-based ETL pipelines were used to process the raw datasets.

The processing stage involved:

- Data Cleaning
- Missing Value Handling
- Duplicate Removal
- Data Validation
- Standardization
- Feature Engineering

Step 3: Database Storage

The cleaned datasets were loaded into a SQLite database for structured storage and analytical querying.

Step 4: Analytics & Visualization

Power BI was connected to the SQLite database and processed datasets to generate interactive dashboards, KPI cards, trend analysis, and business insights.

6.3 Star Schema Design

A star schema architecture was implemented to support efficient analytical queries and reporting.

The schema consists of dimension tables containing descriptive attributes and fact tables containing measurable business metrics.

The star schema design improves query performance and simplifies dashboard development.

Dimension Tables

- dim_fund
- dim_date

Fact Tables

- fact_nav

- fact_transactions
- fact_performance
- fact_aum

The fact tables are linked to dimension tables using primary and foreign key relationships.

6.4 Fact Tables

fact_nav

Stores historical NAV records for mutual fund schemes.

Key Attributes:

- nav_id
- amfi_code
- date_key
- nav

Purpose:

Used for NAV trend analysis, return calculations, CAGR computation, and benchmark comparisons.

fact_transactions

Stores investor transaction information.

Key Attributes:

- transaction_id
- amfi_code
- date_key
- transaction_type
- amount_inr

Purpose:

Used for transaction analysis, investor behavior studies, SIP trend analysis, and demographic reporting.

fact_performance

Stores performance metrics calculated for each mutual fund scheme.

Key Attributes:

- return_1yr_pct
- return_3yr_pct
- return_5yr_pct
- sharpe_ratio
- sortino_ratio
- alpha
- beta
- max_drawdown_pct

Purpose:

Used for fund ranking, risk analysis, and performance evaluation.

fact_aum

Stores Assets Under Management information.

Key Attributes:

- date_key
- aum_value

Purpose:

Used for industry growth analysis and AUM trend reporting.

6.5 Dimension Tables

dim_fund

Contains descriptive information about mutual fund schemes.

Key Attributes:

- amfi_code
- scheme_name
- fund_house
- category
- sub_category

- plan
- benchmark
- expense_ratio

Purpose:

Provides classification and filtering capabilities for analytical reports.

dim_date

Contains calendar-related information.

Key Attributes:

- date_key
- date
- day
- month
- quarter
- year

Purpose:

Supports time-series analysis and date-based filtering across all reports.

6.6 Database Relationships

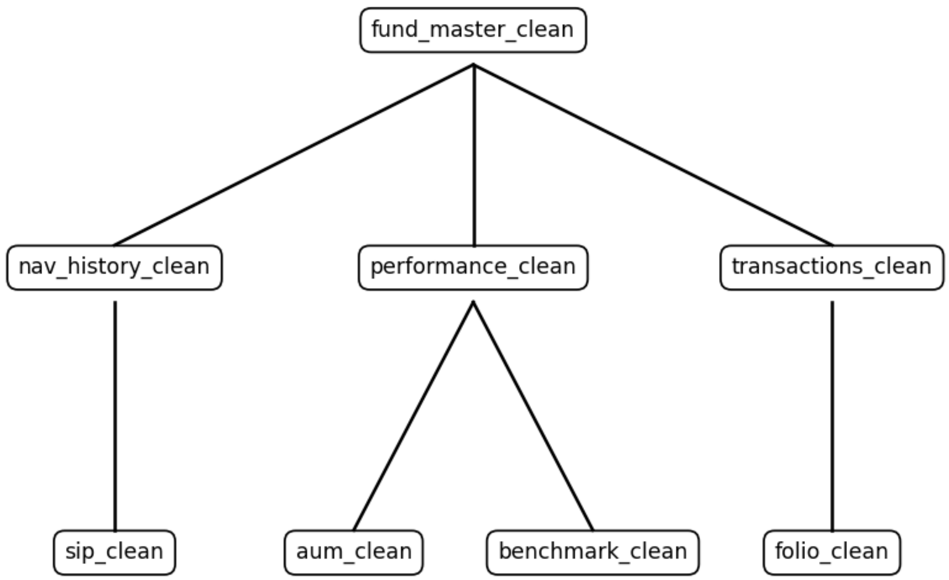
The database was designed using primary and foreign key relationships to maintain data integrity and consistency.

Relationship Structure:

- dim_fund → fact_nav
- dim_fund → fact_transactions
- dim_fund → fact_performance
- dim_date → fact_nav
- dim_date → fact_transactions
- dim_date → fact_aum

This relationship model enables efficient joins and analytical querying across multiple datasets.

Figure 6.2: Database Relationship Model



6.7 Architecture Benefits

The implemented architecture provides several advantages:

- Centralized data storage
- Improved query performance
- Simplified analytical processing
- Better scalability for future enhancements
- Consistent data governance
- Efficient dashboard integration
- Support for advanced analytics and reporting

The architecture serves as the foundation for all analytical operations performed within the Mutual Fund Analytics Platform and ensures reliable delivery of business insights through the Power BI dashboard.

7. ETL Pipeline

7.1 ETL Overview

The ETL (Extract, Transform, Load) pipeline was developed to ensure that raw mutual fund datasets could be converted into a clean, structured, and analysis-ready format. Since the project integrates multiple datasets from different domains of the mutual fund ecosystem, a robust ETL process was essential to maintain data quality and consistency.

The ETL pipeline was implemented using Python and formed the foundation of the overall analytics workflow. It enabled efficient data preparation, validation, and storage for subsequent analysis and visualization.

The ETL process consisted of three primary stages:

- Extract
- Transform
- Load

Each stage played a critical role in preparing the data for analytical processing.

7.2 Data Extraction

The extraction phase involved collecting raw datasets from source files and loading them into Python for processing.

The datasets extracted included:

- Fund Master Dataset
- NAV History Dataset
- Performance Dataset
- Investor Transactions Dataset
- SIP Dataset
- AUM Dataset
- Category Dataset
- Benchmark Dataset
- Folio Dataset

Pandas was used to import the datasets into DataFrames for further processing.

The extraction process included:

- Reading CSV files

- Verifying dataset availability
- Checking schema consistency
- Identifying missing records
- Validating source file structures

The extracted data served as the initial input for transformation and validation activities.

7.3 Data Transformation

The transformation stage focused on improving data quality and preparing datasets for analysis.

Several preprocessing operations were performed during this phase.

Data Cleaning

The following cleaning activities were carried out:

- Removal of duplicate records
- Handling missing values
- Standardization of column names
- Correction of inconsistent values
- Validation of numeric fields
- Formatting of date columns

Data Standardization

To maintain consistency across datasets:

- Date fields were converted into standardized formats.
- Categorical values were normalized.
- Numerical columns were converted to appropriate data types.
- Naming conventions were standardized.

Feature Engineering

Additional analytical fields were created to support reporting and visualization requirements.

Examples include:

- Return calculations
- Risk metrics

- Category aggregations
- Performance indicators
- Date hierarchy attributes

The transformation stage ensured that the datasets were clean, consistent, and suitable for analytical processing.

7.4 Data Loading

After transformation, the processed datasets were loaded into a SQLite database.

SQLite was selected because of its lightweight architecture, ease of integration, and suitability for analytical projects.

The loading process included:

- Database creation
- Table generation
- Data insertion
- Relationship validation
- Row count verification

The cleaned datasets were stored in structured tables that supported both SQL analytics and Power BI reporting.

The database acted as the central repository for all analytical operations performed throughout the project.

7.5 Data Validation

Data validation was performed at multiple stages of the ETL process to ensure reliability and accuracy.

Validation checks included:

Completeness Validation

- Missing value analysis
- Null value detection
- Dataset completeness verification

Consistency Validation

- Column name verification

- Data type validation
- Cross-dataset consistency checks

Accuracy Validation

- Numerical range validation
- Return value verification
- AUM consistency checks
- Transaction amount validation

Integrity Validation

- Primary key validation
- Foreign key relationship verification
- Duplicate record detection

These validation procedures helped maintain high data quality standards throughout the project.

7.6 ETL Challenges

Several challenges were encountered during the ETL process.

Data Quality Issues

Some datasets contained missing values, inconsistent formats, and duplicate records that required additional preprocessing.

Schema Variations

Different datasets followed different structures and naming conventions, requiring schema standardization.

Date Standardization

Date fields across multiple datasets needed to be converted into a consistent format to support time-series analysis.

Data Integration

Combining datasets from multiple domains required careful relationship mapping and validation.

These challenges were addressed through automated cleaning procedures and validation checks.

7.7 ETL Outcomes

The ETL pipeline successfully transformed raw mutual fund datasets into a clean and structured analytical environment.

Key outcomes achieved include:

- Successful ingestion of all project datasets.
- Standardized and validated data structures.
- Cleaned datasets ready for analysis.
- Creation of a centralized SQLite database.
- Improved data quality and consistency.
- Seamless integration with Power BI dashboards.
- Reliable foundation for EDA, performance analytics, and advanced analytics.

The ETL pipeline served as the backbone of the Mutual Fund Analytics Platform and enabled efficient execution of all downstream analytical processes.

8. Data Cleaning & Transformation

8.1 Data Quality Assessment

Before performing analytical operations, a comprehensive assessment of data quality was conducted across all datasets. The objective of this assessment was to identify issues that could affect analytical accuracy and dashboard performance.

The evaluation focused on:

- Missing values
- Duplicate records
- Invalid data types
- Inconsistent formats
- Outliers and anomalies
- Referential integrity issues

Each dataset was reviewed independently to understand its structure, completeness, and reliability before proceeding with data cleaning activities.

8.2 Missing Value Handling

Missing values were identified and analyzed across multiple datasets. Appropriate handling techniques were applied based on the nature and significance of the missing information.

The following approaches were used:

- Forward filling time-series values where appropriate
- Replacing missing categorical values with standardized labels
- Removing records with insufficient information
- Validating critical fields before loading

For the NAV dataset, missing values caused by weekends and market holidays were handled through forward-filling techniques to maintain continuity in time-series analysis.

This process ensured that calculations such as returns, volatility, and performance metrics remained accurate.

8.3 Duplicate Record Removal

Duplicate records can significantly impact analytical results and performance calculations. Therefore, duplicate detection and removal procedures were implemented across all datasets.

The cleaning process involved:

- Identifying exact duplicate rows
- Verifying primary key uniqueness
- Removing redundant observations
- Revalidating row counts after cleaning

Particular attention was given to transaction and NAV datasets, where duplicate entries could distort investment trends and performance calculations.

8.4 Data Standardization

Data standardization was performed to create consistency across datasets and improve compatibility during integration.

The following standardization activities were carried out:

Date Standardization

All date fields were converted into a common datetime format to support time-series analysis and reporting.

Column Name Standardization

Column names were converted into a consistent naming convention to simplify data processing and SQL querying.

Examples include:

- Lowercase formatting
- Underscore-separated naming conventions
- Removal of special characters

Categorical Standardization

Categorical values were normalized to remove inconsistencies.

Examples:

- SIP
- Lumpsum
- Redemption

Standardized values ensured accurate aggregation and reporting.

Numeric Standardization

Numeric columns were validated and converted into appropriate data types to support analytical calculations.

8.5 Data Validation Rules

Validation rules were implemented to ensure data accuracy and business consistency.

NAV Validation

Rules applied:

- NAV values must be greater than zero.
- Invalid or negative NAV records were flagged.
- Date continuity was verified.

Transaction Validation

Rules applied:

- Transaction amount must be positive.
- Transaction types must belong to predefined categories.
- Invalid transaction records were flagged for review.

Performance Validation

Rules applied:

- Return values must be numeric.
- Expense ratios must fall within acceptable ranges.
- Risk metrics must be logically consistent.

Database Validation

Rules applied:

- Primary key uniqueness checks
- Foreign key integrity checks
- Row count verification after loading

These validation procedures improved data reliability and reduced analytical errors.

8.6 Feature Engineering

Additional fields and derived metrics were created to support analytical and reporting requirements.

The feature engineering process included:

Time-Based Features

- Year
- Quarter
- Month
- Day

Performance Metrics

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Analytical Attributes

- Risk Grade
- Category Aggregations
- Investor Segmentation
- Transaction Classifications

These engineered features enhanced analytical depth and enabled advanced reporting capabilities.

8.7 Transformation Outcomes

The data cleaning and transformation process successfully converted raw datasets into structured and analysis-ready assets.

The major outcomes achieved include:

- Improved data accuracy and consistency.
- Removal of duplicate and invalid records.
- Standardized dataset structures.
- Enhanced data quality through validation checks.
- Creation of derived analytical metrics.
- Improved compatibility with SQLite and Power BI.
- Reliable foundation for exploratory analysis and dashboard development.

The transformed datasets were subsequently used for database design, exploratory data analysis, performance analytics, advanced analytics, and interactive dashboard creation throughout the project.

9. Exploratory Data Analysis (EDA)

9.1 EDA Overview

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, and underlying patterns present within the mutual fund datasets. The primary objective of EDA was to transform raw data into meaningful insights by identifying trends, distributions, relationships, and anomalies across various dimensions of the mutual fund ecosystem.

The analysis focused on industry growth, fund performance, investor participation, SIP trends, category-level activity, benchmark movements, and transaction behavior. The insights generated during this stage played a crucial role in dashboard design and business intelligence reporting.

EDA was conducted using Python libraries such as Pandas, NumPy, Matplotlib, and Plotly.

9.2 Industry AUM Analysis

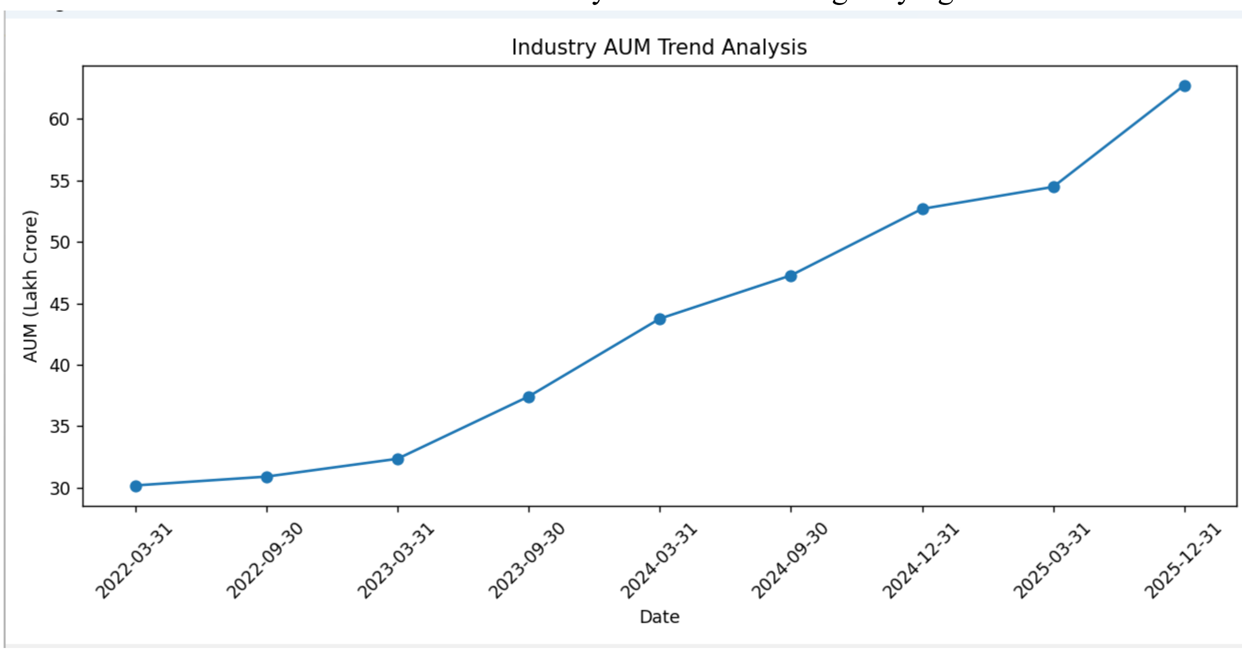
Assets Under Management (AUM) is one of the most important indicators of growth in the mutual fund industry. The AUM dataset was analyzed to understand industry expansion over time and identify trends in asset accumulation.

The analysis focused on:

- Total Industry AUM Growth
- Year-wise AUM Trends
- Monthly AUM Movements
- Asset Growth Patterns

The findings indicated a strong upward trend in total AUM, reflecting increasing investor participation and growing confidence in mutual fund investments.

The AUM trend also demonstrated the industry's resilience during varying market conditions.



9.3 SIP Growth Analysis

Systematic Investment Plans (SIPs) have become a major contributor to mutual fund investments in India. The SIP dataset was analyzed to evaluate investor participation through periodic investments.

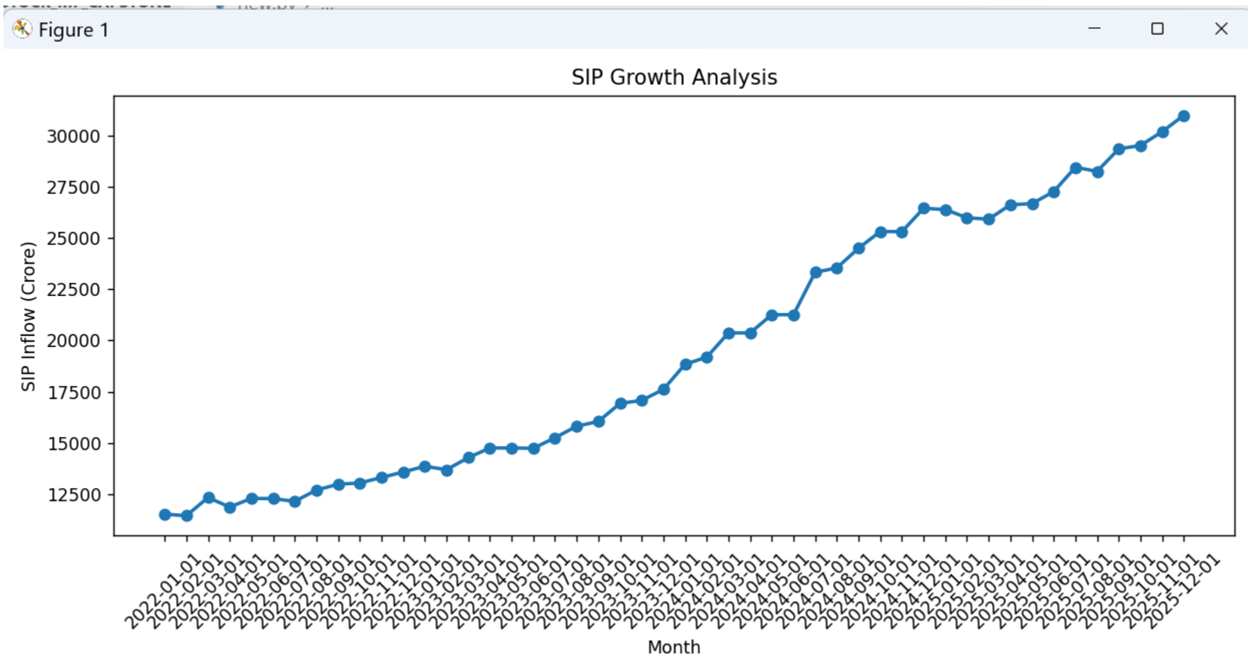
Key areas of analysis included:

- SIP Inflow Trends
- Active SIP Accounts

- SIP Asset Growth
- Monthly SIP Contributions

The analysis revealed consistent growth in SIP inflows, indicating increasing adoption of disciplined investing practices among retail investors.

SIP activity remained strong throughout the observed period, highlighting investor confidence in long-term wealth creation.



9.4 Folio Growth Analysis

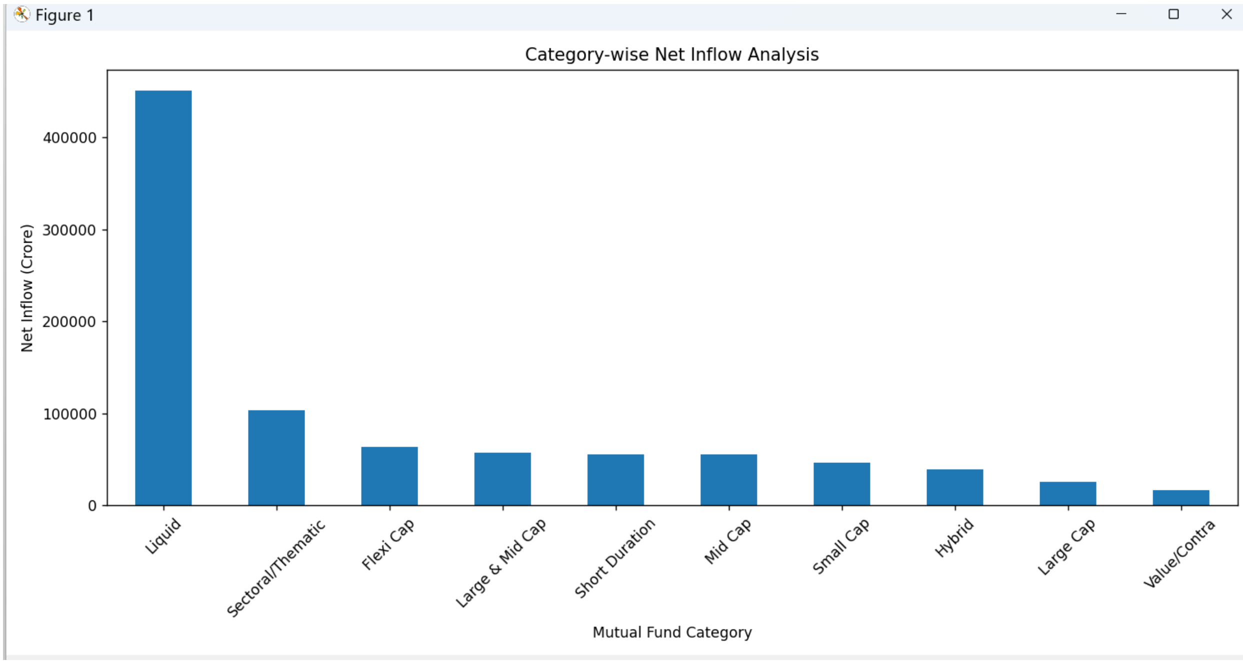
Folio data was analyzed to understand the growth of investor participation within the mutual fund industry.

The analysis included:

- Total Folio Growth
- Year-wise Folio Trends
- Investor Participation Patterns

The increasing number of folios reflected growing awareness of mutual fund investments and broader participation across investor segments.

The results suggested that mutual funds continue to attract new investors while retaining existing participants.



9.5 Category-wise Analysis

Category-level analysis was performed to identify investor preferences and understand fund flow behavior across different mutual fund segments.

Categories analyzed included:

- Large Cap Funds
- Mid Cap Funds
- Small Cap Funds
- Flexi Cap Funds
- Hybrid Funds
- Liquid Funds
- Sectoral/Thematic Funds

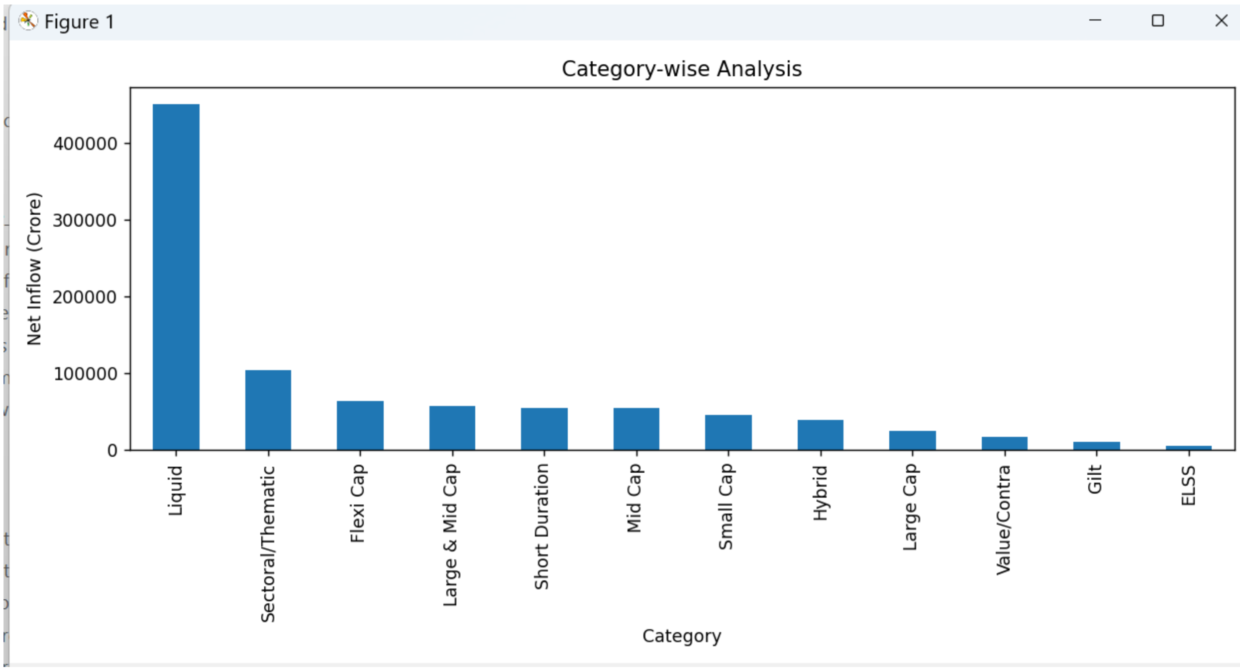
The analysis focused on:

- Category-wise Inflows
- Category-wise AUM
- Relative Popularity

- Growth Trends

Certain categories demonstrated significantly higher inflows compared to others, indicating changing investor preferences and market sentiment.

The category analysis provided valuable insights into sector-specific investment trends.



9.6 Benchmark Analysis

Benchmark indices were analyzed to understand overall market performance and provide context for mutual fund evaluations.

The benchmark dataset included major market indices such as:

- Nifty 50
- Nifty 100

The analysis examined:

- Index Movement Trends
- Market Growth Patterns
- Benchmark Performance Comparisons

Benchmark analysis served as a reference point for evaluating fund performance and measuring relative returns.

The results highlighted periods of market growth, volatility, and recovery throughout the analysis period.

9.7 Investor Transaction Analysis

Investor transaction data was analyzed to understand investment behavior, demographic trends, and geographic participation patterns.

The analysis included:

Transaction Type Analysis

Evaluation of:

- SIP Investments
- Lumpsum Investments
- Redemptions

Geographic Analysis

Analysis of:

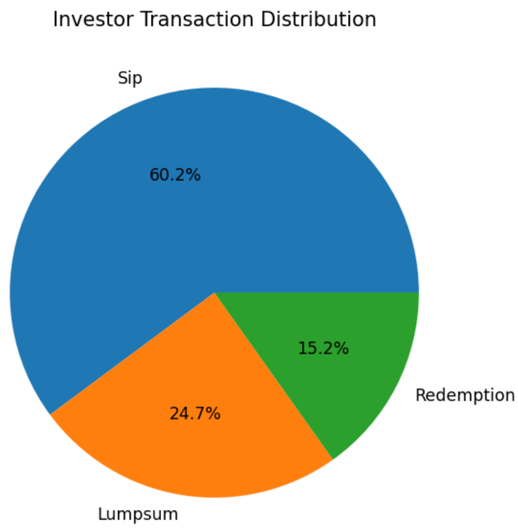
- State-wise Transactions
- City Tier Distribution
- Regional Participation

Demographic Analysis

Analysis of:

- Age Groups
- Income Segments
- Investor Categories

The results provided valuable insights into investor preferences, transaction patterns, and participation across different demographic segments.



9.8 Key EDA Findings

The Exploratory Data Analysis phase generated several important insights:

- The mutual fund industry demonstrated strong growth in Assets Under Management (AUM).
- SIP inflows showed consistent upward momentum throughout the analysis period.
- Investor participation increased steadily as reflected in folio growth.
- Category-wise inflows revealed distinct investor preferences across fund segments.
- Benchmark indices exhibited long-term growth despite short-term fluctuations.
- Transaction analysis highlighted significant regional and demographic participation patterns.
- Multiple datasets exhibited strong potential for performance evaluation and advanced analytics.

The insights generated during EDA provided the analytical foundation for subsequent performance analysis, advanced analytics, and dashboard development activities.

10. Performance Analytics

10.1 Overview

Performance Analytics was conducted to evaluate mutual fund schemes using both return-based and risk-adjusted performance metrics. While absolute returns provide information about fund growth, they do not fully capture the level of risk undertaken to achieve those returns.

To provide a comprehensive evaluation framework, multiple performance indicators were calculated and analyzed. These metrics enabled comparison of funds across different categories while considering risk, volatility, and benchmark performance.

The analysis focused on the following metrics:

- Compound Annual Growth Rate (CAGR)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Benchmark Comparison

These measures collectively provided a balanced assessment of fund performance.

Metric	Description	Purpose
CAGR	Compound Annual Growth Rate	Measures annualized fund growth
Sharpe Ratio	Risk-adjusted return metric	Evaluates return per unit of risk
Sortino Ratio	Downside risk-adjusted return	Focuses on harmful volatility
Alpha	Excess return over benchmark	Measures fund manager performance
Beta	Market sensitivity indicator	Measures systematic risk
Maximum Drawdown	Largest peak-to-trough decline	Evaluates downside risk
VaR	Value at Risk	Estimates potential loss
CVaR	Conditional Value at R	Measures extreme downside risk

\

10.2 CAGR Analysis

Compound Annual Growth Rate (CAGR) was used to measure the annualized growth rate of mutual fund schemes over different investment horizons.

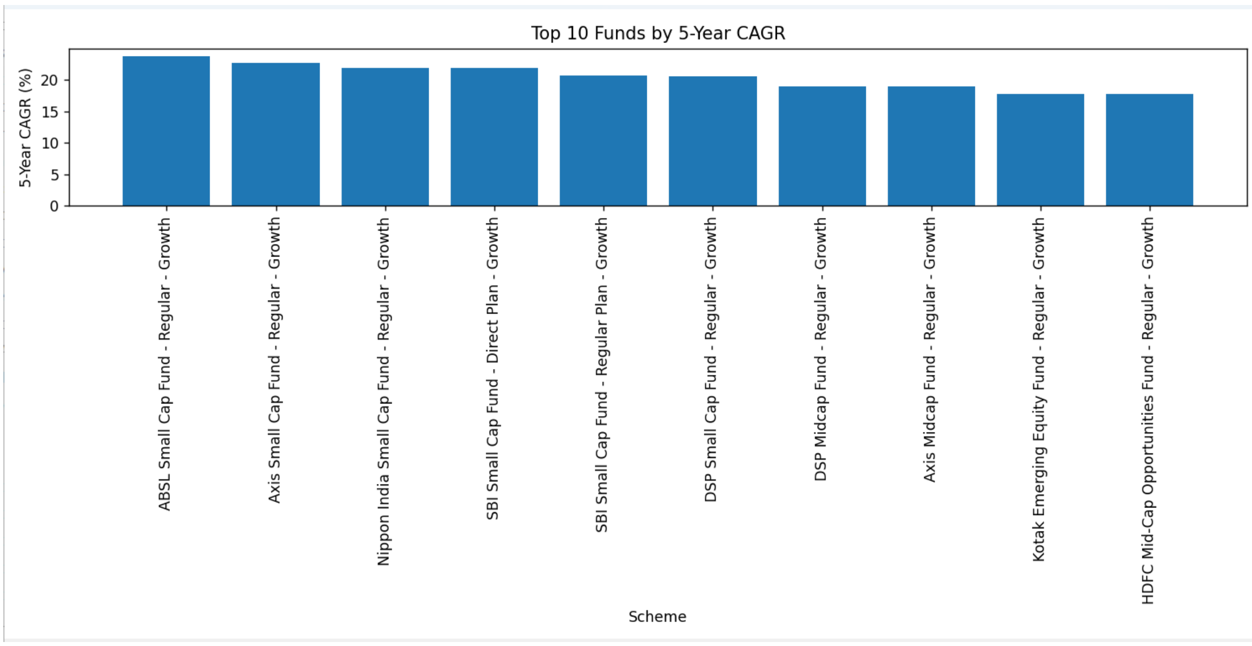
CAGR was calculated for:

- 1-Year Period
- 3-Year Period
- 5-Year Period

The metric provides a standardized measure of growth by smoothing fluctuations and representing the average annual return generated over a specified period.

The analysis enabled comparison of long-term performance across multiple schemes and categories.

Funds with consistently strong CAGR values demonstrated superior wealth creation potential over extended investment horizons.



10.3 Sharpe Ratio Analysis

The Sharpe Ratio was calculated to evaluate risk-adjusted returns.

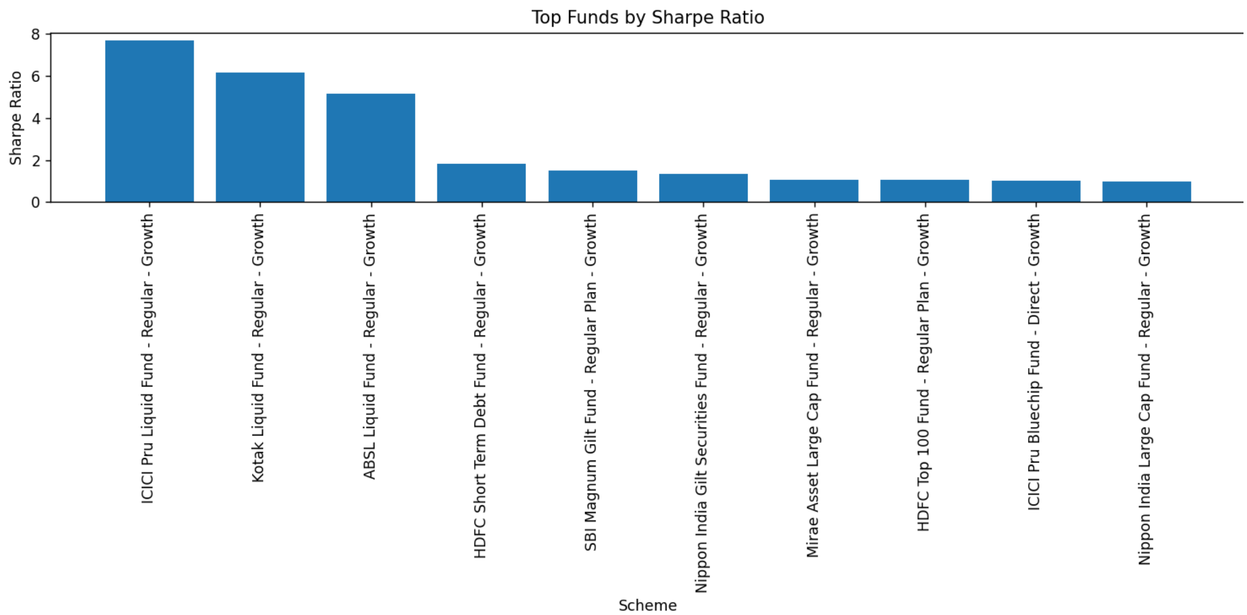
The metric measures the amount of excess return generated for each unit of total risk undertaken by a fund.

A higher Sharpe Ratio indicates:

- Better risk-adjusted performance
- More efficient portfolio management
- Superior return generation relative to volatility

The analysis identified funds that delivered strong returns while maintaining controlled levels of risk.

Sharpe Ratio rankings were used to compare the efficiency of different mutual fund schemes.



10.4 Sortino Ratio Analysis

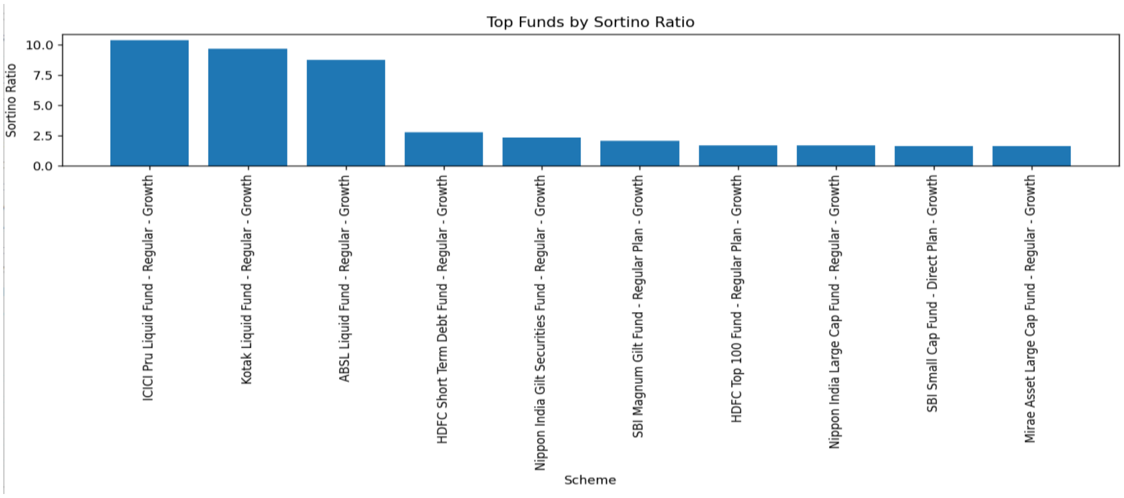
The Sortino Ratio was used as an alternative risk-adjusted performance measure.

Unlike the Sharpe Ratio, which considers total volatility, the Sortino Ratio focuses only on downside risk.

This metric provides a more accurate assessment of investment performance by penalizing harmful volatility while ignoring positive fluctuations.

The analysis highlighted funds that effectively minimized downside risk while maintaining attractive returns.

Funds with higher Sortino Ratios demonstrated stronger downside protection and more stable performance characteristics.



10.5 Alpha and Beta Analysis

Alpha and Beta were calculated to assess the relationship between mutual fund performance and market benchmarks.

Alpha Analysis

Alpha measures a fund manager's ability to generate returns beyond those expected from market movements.

A positive Alpha indicates:

- Outperformance relative to the benchmark
- Value creation through active management

The analysis identified schemes that consistently generated excess returns beyond benchmark expectations.

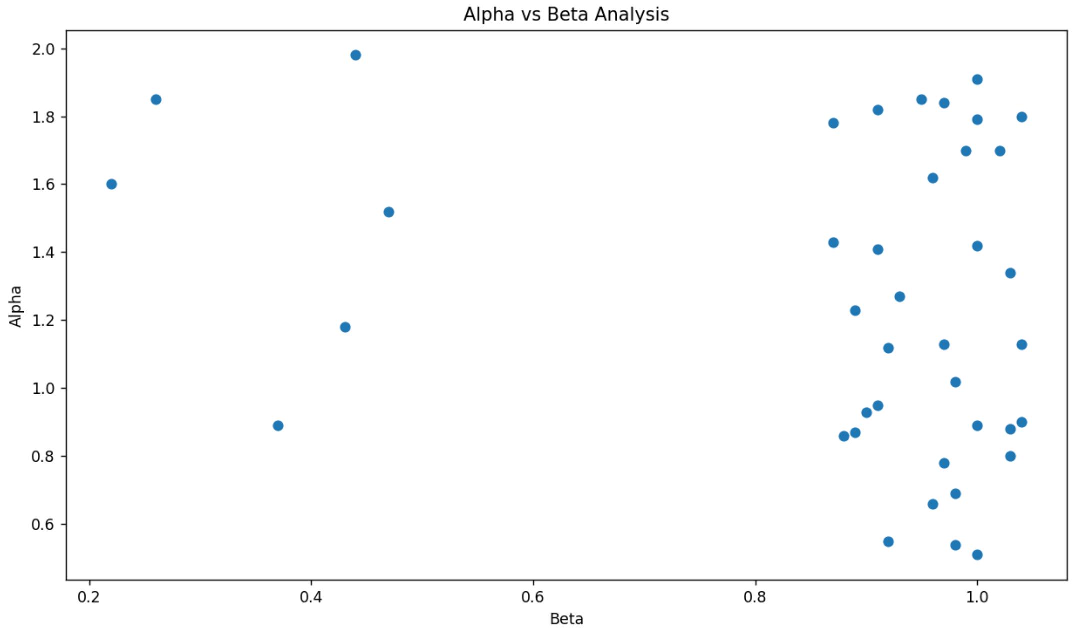
Beta Analysis

Beta measures the sensitivity of a fund's returns relative to market movements.

Interpretation of Beta values:

- Beta > 1 : More volatile than the market
- Beta = 1 : Similar volatility as the market
- Beta < 1 : Less volatile than the market

Beta analysis provided insights into fund risk characteristics and market responsiveness.



10.6 Maximum Drawdown Analysis

Maximum Drawdown was calculated to evaluate downside risk and capital preservation capabilities.

The metric measures the largest decline experienced by a fund from its peak value before recovery.

The analysis focused on:

- Magnitude of losses
- Recovery characteristics
- Downside risk exposure

Funds with lower maximum drawdowns demonstrated stronger resilience during market corrections and periods of volatility.

Maximum Drawdown served as an important indicator for risk-conscious investors.

10.7 Fund Ranking Methodology

A comparative ranking framework was developed to evaluate mutual fund schemes across multiple performance dimensions.

The ranking methodology considered:

- Return Performance

- Risk-Adjusted Returns
- Alpha Generation
- Downside Risk Characteristics
- Benchmark Outperformance

Each metric contributed to a comprehensive assessment of overall fund quality.

The ranking process enabled identification of top-performing schemes based on both return potential and risk management effectiveness.

scheme_name	return_5yr_pct	sharpe_ratio	sortino_ratio	alpha	beta
ABSL Small Cap Fund - Regular - Growth	23.80	0.90	1.47	1.84	0.97
Axis Small Cap Fund - Regular - Growth	22.62	0.84	1.40	0.51	1.00
Nippon India Small Cap Fund - Regular - Growth	21.88	0.81	1.14	0.80	1.03
SBI Small Cap Fund - Direct Plan - Growth	21.82	0.93	1.67	1.13	1.04
SBI Small Cap Fund - Regular Plan - Growth	20.67	0.94	1.35	1.23	0.89
DSP Small Cap Fund - Regular - Growth	20.61	0.80	1.23	0.69	0.98
DSP Midcap Fund - Regular - Growth	19.00	0.90	1.50	1.02	0.98
Axis Midcap Fund - Regular - Growth	18.94	0.80	1.18	1.42	1.00
Kotak Emerging Equity Fund - Regular - Growth	17.75	0.96	1.27	1.91	1.00
HDFC Mid-Cap Opportunities Fund - Regular - Growth	17.69	0.87	1.44	0.95	0.91

10.8 Benchmark Comparison

Mutual fund performance was compared against benchmark indices to evaluate relative effectiveness.

The benchmark analysis included:

- Nifty 50
- Nifty 100

Comparisons focused on:

- Return Performance
- Relative Growth
- Risk Characteristics
- Tracking Behavior

Benchmark comparison provided context for evaluating whether a fund was generating value beyond general market performance.

The analysis helped identify funds capable of consistently outperforming market indices.

10.9 Key Performance Insights

The Performance Analytics phase generated several important findings.

Key observations include:

- Significant differences existed between absolute returns and risk-adjusted returns.
- Several funds delivered superior performance when evaluated using Sharpe and Sortino Ratios.
- Alpha analysis identified funds that consistently outperformed benchmark expectations.
- Beta values revealed varying levels of market sensitivity across fund categories.
- Maximum Drawdown analysis highlighted differences in downside risk exposure.
- Benchmark comparisons provided a clear understanding of relative fund performance.
- Risk-adjusted metrics proved essential for identifying sustainable long-term performers.

The insights generated during this phase formed the basis for advanced analytics, fund comparison, dashboard development, and business intelligence reporting.

11. Advanced Analytics

11.1 Overview

Advanced Analytics was performed to extend the scope of traditional performance analysis and generate deeper insights into mutual fund behavior, risk exposure, investor participation, and portfolio characteristics. While conventional metrics such as CAGR and Sharpe Ratio provide valuable information, advanced analytical techniques enable a more detailed assessment of investment risk, stability, diversification, and investor engagement.

The objective of this phase was to apply quantitative methods that support better understanding of fund performance under different market conditions and improve the overall analytical capability of the platform.

The analysis focused on:

- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Fund Recommendation Framework
- Concentration Analysis (HHI)

11.2 Value at Risk (VaR)

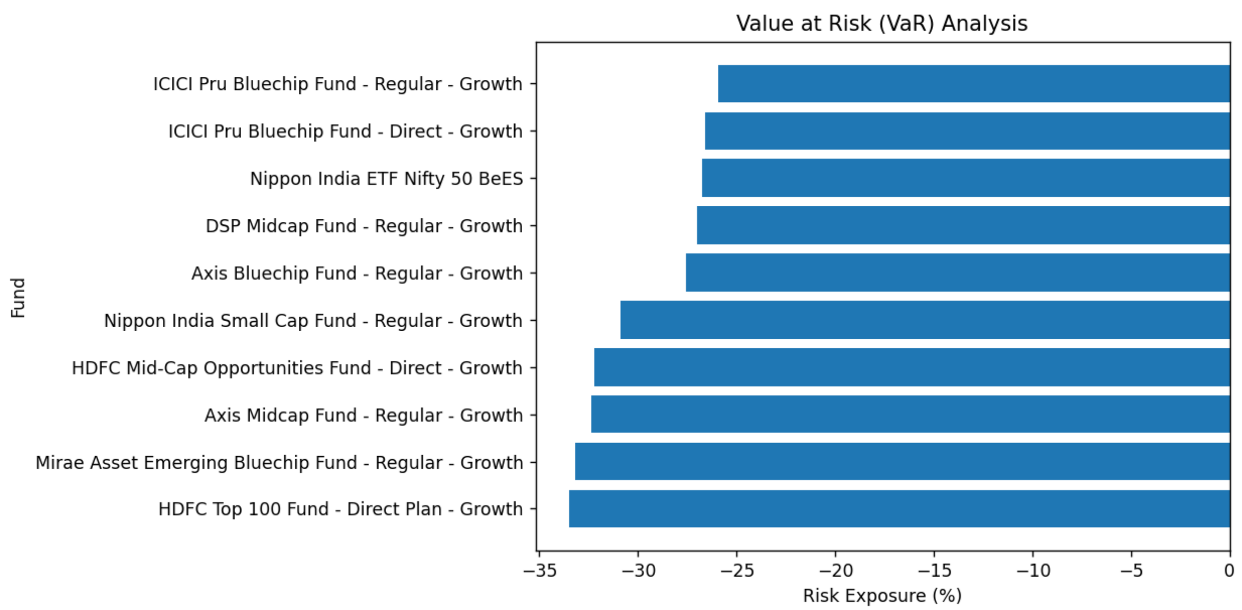
Value at Risk (VaR) was used to estimate the potential loss that a mutual fund could experience over a specified period under normal market conditions.

11.2.1 VaR Formula

[
VaR = \mu - Z\sigma
]

Where:

- (\mu) = Mean Return
- (Z) = Confidence Level Z-Score
- (\sigma) = Standard Deviation of Returns



11.2.2 Interpretation

VaR provides an estimate of the maximum expected loss at a specific confidence level over a defined time period.

The metric helps:

- Measure downside risk.
- Compare risk exposure across funds.
- Support risk management decisions.

Funds with lower VaR values generally exhibit lower short-term downside risk.

11.3 Conditional Value at Risk (CVaR)

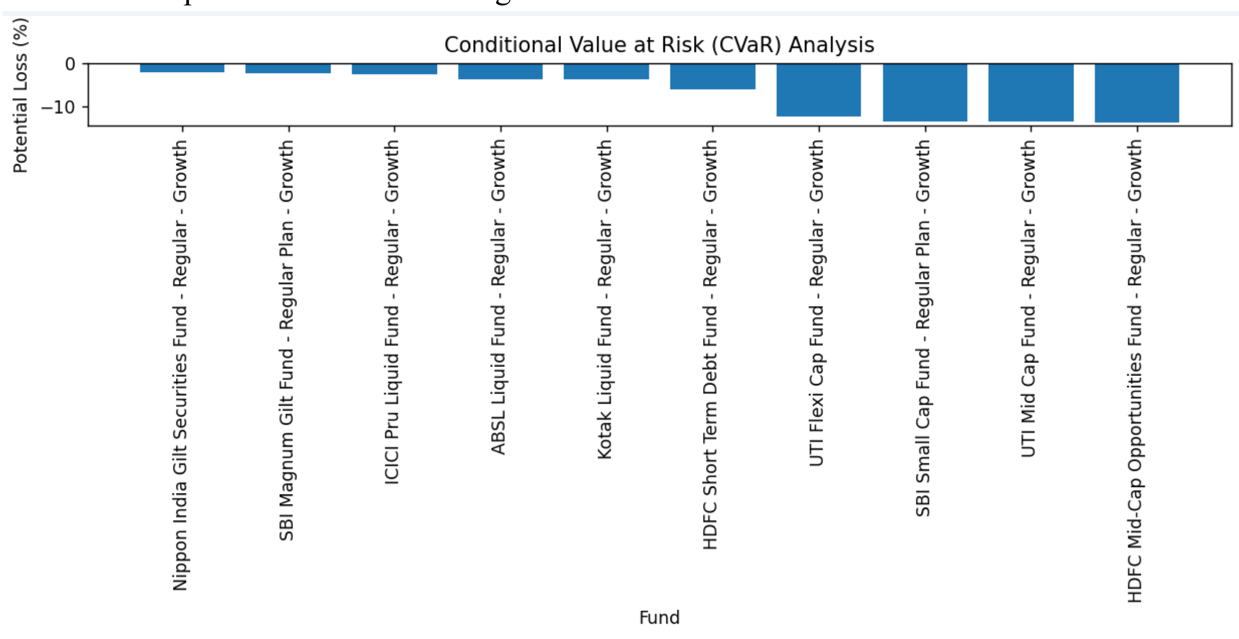
Conditional Value at Risk (CVaR), also known as Expected Shortfall, was calculated to measure the average loss occurring beyond the VaR threshold.

11.3.1 CVaR Formula

[
CVaR = E(Loss \mid Loss > VaR)
]

Where:

- Loss > VaR represents losses exceeding the VaR threshold.



11.3.2 Interpretation

CVaR provides a more comprehensive view of extreme downside risk compared to VaR.

The metric helps:

- Evaluate tail-risk exposure.
- Measure potential losses during severe market conditions.
- Improve risk assessment beyond standard volatility measures.

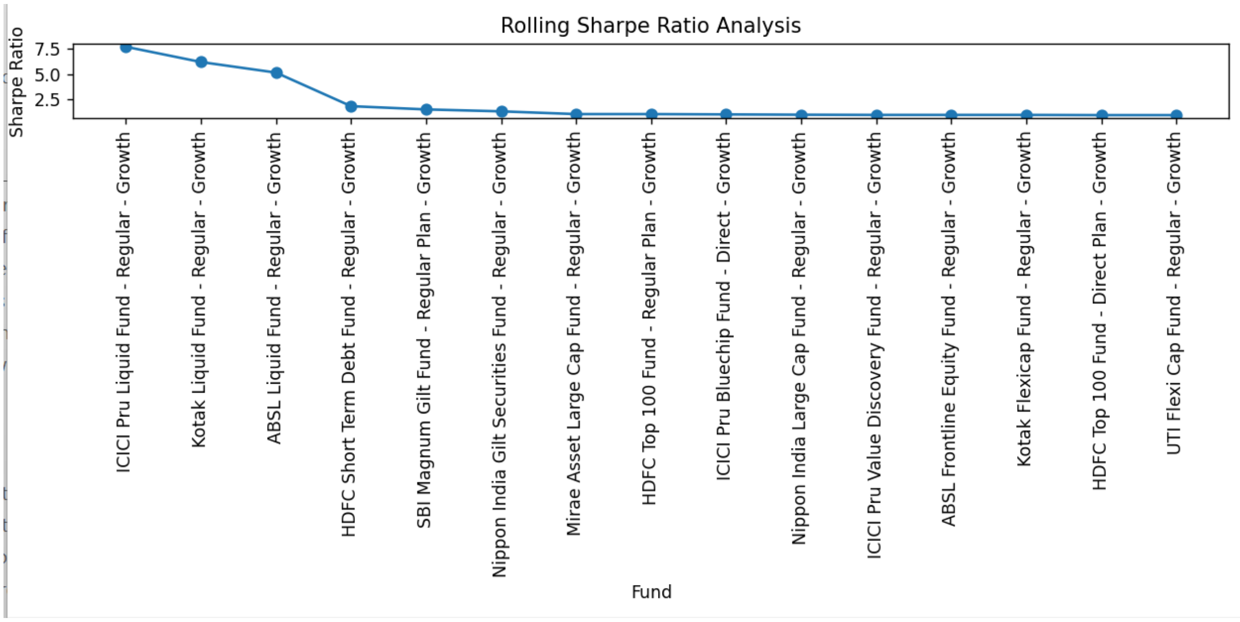
Lower CVaR values indicate stronger protection against extreme losses.

11.4 Rolling Sharpe Ratio Analysis

Rolling Sharpe Ratio analysis was conducted to evaluate the consistency of risk-adjusted performance over time.

11.4.1 Rolling Sharpe Formula

[
Rolling\ Sharpe =
\frac{Rolling\ Return - Risk\ Free\ Rate}
{Rolling\ Standard\ Deviation}
]



11.4.2 Interpretation

Unlike a single Sharpe Ratio calculation, rolling analysis evaluates performance across multiple time windows.

The analysis helps:

- Identify performance stability.
- Detect changing risk-return characteristics.
- Compare consistency among funds.

Funds exhibiting stable rolling Sharpe values demonstrate more reliable long-term performance.

11.5 Investor Cohort Analysis

Investor Cohort Analysis was performed to understand participation patterns among different investor groups.

Investors were segmented using:

- Age Group
- Income Category
- State
- City Tier
- Investment Type

11.5.1 Objectives

The analysis was conducted to:

- Identify dominant investor segments.
- Study behavioral patterns.
- Understand participation trends.
- Compare investment activity across demographic groups.

11.5.2 Interpretation

The analysis revealed variations in investment preferences across demographic segments and highlighted differences in participation levels among investor groups.

11.6 SIP Continuity Analysis

SIP Continuity Analysis was performed to evaluate the persistence of systematic investment behavior over time.

11.6.1 Metrics Evaluated

- Active SIP Accounts
- Monthly SIP Contributions
- SIP Growth Trends
- Participation Retention

11.6.2 Interpretation

Consistent SIP contributions indicate investor confidence and long-term investment discipline.

The analysis provided insights into:

- Investor retention
- Participation stability
- Long-term investment behavior

Strong SIP continuity generally reflects healthy growth within the mutual fund industry.

11.7 Fund Recommendation Framework

A rule-based recommendation framework was developed to identify high-performing mutual funds based on multiple performance indicators.

11.7.1 Evaluation Criteria

The framework considered:

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Benchmark Outperformance

11.7.2 Methodology

Each fund was evaluated using a weighted scoring approach.

Funds achieving strong scores across both return and risk dimensions received higher rankings.

11.7.3 Interpretation

The framework provides a structured and objective method for identifying quality mutual fund schemes based on historical performance characteristics.

11.8 Concentration Analysis (HHI)

The Herfindahl-Hirschman Index (HHI) was used to measure concentration levels across fund categories and investment distributions.

11.8.1 HHI Formula

$$
HHI = \sum s_i^2
$$

Where:

- (s_i) = Market Share of Category (i)

11.8.2 Interpretation

HHI measures the degree of concentration within a market or investment distribution.

Interpretation:

- Low HHI → High Diversification
- Moderate HHI → Balanced Concentration
- High HHI → Significant Concentration Risk

The analysis helps identify whether investments are spread across multiple categories or concentrated in a few segments.

11.9 Advanced Analytics Insights

Several important insights emerged from the advanced analytics phase.

Key findings include:

- Risk exposure varied significantly across mutual fund categories.
- VaR and CVaR provided valuable measures of downside risk.
- Rolling Sharpe analysis revealed differences in performance consistency.
- Investor participation patterns differed across age groups, income levels, and geographic regions.
- SIP contributions demonstrated strong continuity and long-term participation trends.
- Multi-factor evaluation improved fund comparison and ranking accuracy.
- Concentration analysis highlighted the importance of diversification in investment decision-making.

The Advanced Analytics phase strengthened the overall analytical depth of the project and provided meaningful insights beyond traditional performance evaluation metrics.

12. Database Design & SQL Analytics

12.1 Database Overview

A structured relational database was designed to store, manage, and analyze the processed mutual fund datasets. The database serves as the central repository for all analytical operations performed throughout the project.

SQLite was selected as the database management system due to its lightweight architecture, ease of deployment, and seamless integration with Python and Power BI.

The database enables efficient storage, retrieval, and analysis of large volumes of mutual fund data while maintaining consistency and integrity across multiple datasets.

The primary objectives of the database design were:

- Centralized data storage
- Efficient analytical querying
- Data consistency and integrity
- Dashboard integration
- Support for business intelligence reporting

Table 12.1: Database Tables

Table Name	Records	Description
fund_master_clean	40	Master information of mutual fund schemes
nav_history_clean	46,000	Historical NAV records
performance_clean	40	Performance and risk metrics
transactions_clean	32,778	Investor transaction records
sip_clean	48	SIP statistics and inflows
aum_clean	90	Industry AUM information
category_clean	144	Category-wise inflow data
benchmark_clean	8,050	Benchmark index history
folio_clean	21	Investor folio statistics

12.2 Database Schema Design

The database schema was designed using a relational approach to establish meaningful relationships between datasets.

The schema integrates multiple domains of the mutual fund ecosystem, including:

- Fund Information
- NAV History
- Performance Metrics
- Investor Transactions
- SIP Statistics
- Industry AUM Data
- Category Inflows
- Benchmark Indices
- Folio Statistics

The schema structure was designed to minimize redundancy while maintaining efficient access to analytical data.

Database Architecture

The database consists of multiple interconnected tables that collectively support reporting and analytics requirements.

Major tables include:

- fund_master_clean
- nav_history_clean
- performance_clean
- transactions_clean
- sip_clean
- aum_clean
- category_clean
- benchmark_clean
- folio_clean

12.3 Table Structure

Table 12.2: Database Relationships

Parent Table	Child Table	Relationship Key
fund_master_clean	nav_history_clean	amfi_code
fund_master_clean	performance_clean	amfi_code
fund_master_clean	transactions_clean	amfi_code
benchmark_clean	performance_clean	benchmark/index reference
aum_clean	category_clean	date/month
sip_clean	category_clean	month
folio_clean	aum_clean	date

Relationship Summary

```
fund_master_clean
        |
   ┌────┼───────┐
   |    |       |
NAV  Performance Transactions
```

fund_master_clean

Stores scheme-level information.

Key fields:

- amfi_code
- scheme_name
- fund_house
- category
- sub_category
- plan
- benchmark

- expense_ratio_pct
- risk_category

Purpose:

Acts as the master reference table for all mutual fund schemes.

---

nav_history_clean

Stores historical NAV records.

Key fields:

- amfi_code
- date
- nav

Purpose:

Supports NAV trend analysis and return calculations.

---

performance_clean

Stores performance and risk metrics.

Key fields:

- return_1yr_pct
- return_3yr_pct
- return_5yr_pct
- sharpe_ratio
- sortino_ratio
- alpha
- beta
- max_drawdown_pct

Purpose:

Supports performance evaluation and fund ranking.

---

transactions_clean

Stores investor transaction data.

Key fields:

- transaction_date
- transaction_type
- amount_inr
- state
- city_tier
- age_group
- annual_income_lakh

Purpose:

Supports investor behavior and transaction analytics.

---

sip_clean

Stores SIP-related statistics.

Key fields:

- month
- sip_inflow_crore
- active_sip_accounts_lakh
- sip_aum_lakh_crore

Purpose:

Supports SIP growth and participation analysis.

---

aum_clean

Stores industry AUM information.

Key fields:

- date
- total_aum_lakh_crore

Purpose:

Supports industry growth analysis.

---

category_clean

Stores category-wise fund flow information.

Key fields:

- category
- net_inflow_crore

Purpose:

Supports category-level trend analysis.

---

benchmark_clean

Stores benchmark index data.

Key fields:

- date
- index_name
- close_value

Purpose:

Supports benchmark comparison analysis.

---

folio_clean

Stores investor folio statistics.

Key fields:

- date
- folios_crore

Purpose:

Supports investor participation analysis.

12.4 Primary and Foreign Keys

Relationships were established to ensure data consistency and support analytical operations.

Primary Key

The AMFI Code serves as the primary identifier for mutual fund schemes.

Example:

amfi_code

Foreign Key Relationships

Key relationships include:

fund_master_clean → nav_history_clean

fund_master_clean → performance_clean

fund_master_clean → transactions_clean

Date-based relationships include:

DimDate → nav_history_clean

DimDate → aum_clean

DimDate → sip_clean

DimDate → benchmark_clean

DimDate → category_clean

DimDate → folio_clean

These relationships enable efficient joins and dashboard filtering.

12.5 SQL Queries and Analysis

SQL was used extensively to generate analytical insights and support dashboard KPIs.

Total Schemes

SELECT COUNT(*) AS total_schemes

FROM fund_master_clean;

Purpose:

Calculates the total number of mutual fund schemes.

Average Fund Return

SELECT AVG(return_5yr_pct)

FROM performance_clean;

Purpose:

Calculates average long-term fund performance.

Top Performing Funds

SELECT scheme_name,

return_5yr_pct

FROM performance_clean

ORDER BY return_5yr_pct DESC

LIMIT 10;

Purpose:

Identifies top-performing funds.

State-wise Transaction Analysis

SELECT state,

SUM(amount_inr) AS total_amount

FROM transactions_clean

GROUP BY state

ORDER BY total_amount DESC;

Purpose:

Analyzes geographic investment activity.

Category-wise Inflow Analysis

```
SELECT category,
    SUM(net_inflow_crore) AS inflow
FROM category_clean
GROUP BY category
ORDER BY inflow DESC;
```

Purpose:

Identifies popular investment categories.

12.6 Business Insights Generated

Several business insights were generated using SQL-based analysis.

Key insights include:

- Identification of top-performing mutual fund schemes.
- Analysis of investor participation across states.
- Measurement of industry growth through AUM trends.
- Evaluation of SIP contribution patterns.
- Identification of high-growth fund categories.
- Benchmark comparison across market conditions.
- Monitoring of investor folio growth.

These insights directly contributed to dashboard development and reporting.

12.7 Database Performance

The database was optimized to support analytical workloads and dashboard responsiveness.

Performance considerations included:

- Structured schema design
- Optimized joins

- Consistent data types
- Efficient aggregation queries
- Reduced data redundancy

The resulting database provided reliable performance for both SQL analytics and Power BI integration.

12.8 Key Outcomes

The database implementation successfully achieved the following outcomes:

- Centralized storage of all processed datasets.
- Improved analytical efficiency.
- Consistent data management.
- Reliable SQL-based insight generation.
- Seamless Power BI integration.
- Support for advanced analytics and reporting.

The database layer serves as the analytical backbone of the Mutual Fund Analytics Platform and enables efficient delivery of business intelligence insights across all project components.

13. Power BI Dashboard Development

13.1 Dashboard Overview

An interactive Power BI dashboard was developed to transform analytical results into meaningful visual insights. The dashboard serves as the final presentation layer of the Mutual Fund Analytics Platform and enables users to explore industry trends, fund performance, investor behavior, and market movements through an intuitive interface.

The dashboard was designed with a business intelligence approach, focusing on clarity, usability, and interactivity. Multiple datasets were integrated through a structured data model, allowing users to analyze information across different dimensions and time periods.

The dashboard consists of four primary pages:

- Industry Overview
- Fund Performance Analysis
- Investor Analytics

- SIP & Market Trends

Each page focuses on a specific analytical domain and provides actionable insights through visualizations and KPIs.

13.2 Industry Overview Page

The Industry Overview page provides a high-level summary of the mutual fund industry's growth and performance.

Key Performance Indicators (KPIs)

The page includes the following KPI cards:

- Total Assets Under Management (AUM)
- Total SIP Inflows
- Total Folios
- Total Schemes

These KPIs provide a quick overview of industry size and growth.

Visualizations

AUM Trend Analysis

A line chart was used to visualize industry AUM growth over time.

Purpose:

- Monitor industry expansion
- Identify long-term growth patterns
- Evaluate market participation

AUM by Asset Management Company (AMC)

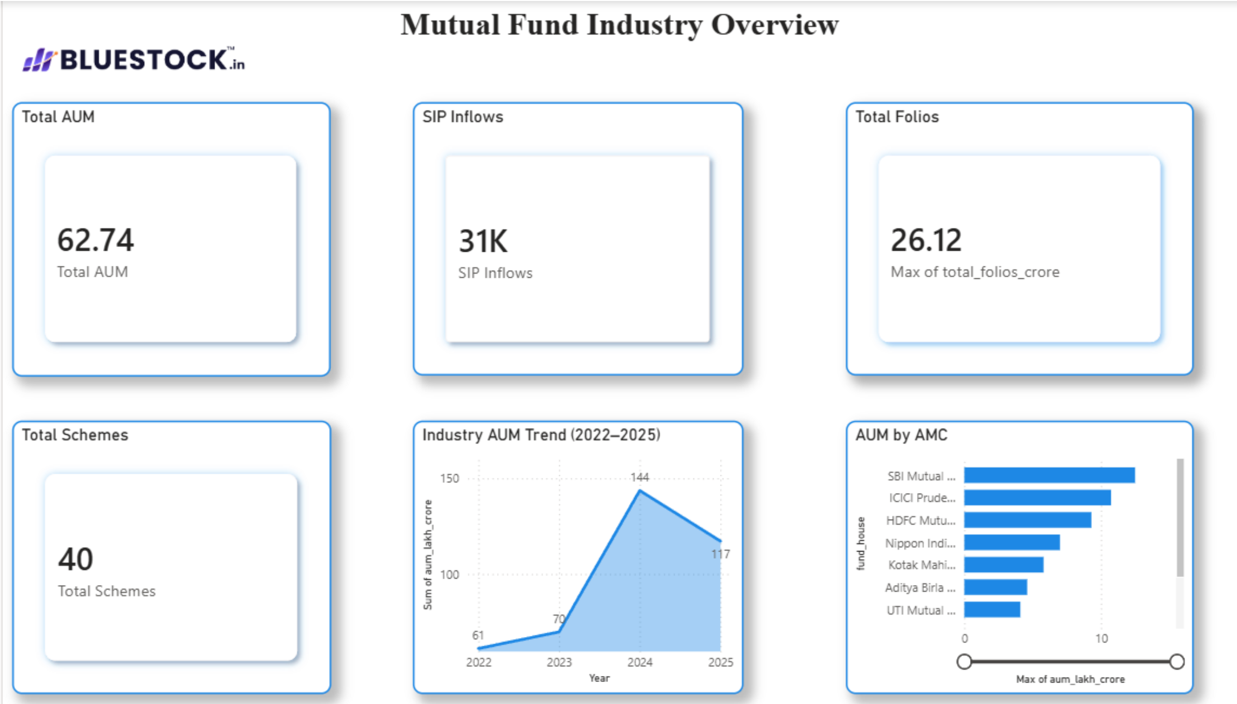
A bar chart was used to compare AUM across different fund houses.

Purpose:

- Compare AMC market share
- Identify leading fund houses
- Evaluate industry concentration

Business Value

The Industry Overview page provides a consolidated snapshot of mutual fund industry performance and serves as the dashboard's executive summary.



13.3 Fund Performance Page

The Fund Performance page focuses on evaluating mutual fund schemes using return and risk metrics.

Risk vs Return Scatter Plot

A scatter plot was developed using:

- Return as the X-axis
- Risk (Standard Deviation) as the Y-axis
- AUM as Bubble Size

Purpose:

- Compare fund performance
- Identify efficient funds
- Visualize risk-return relationships

Fund Scorecard Table

A detailed scorecard was created containing:

- Scheme Name
- Fund House
- Category
- Risk Grade
- Returns
- Sharpe Ratio
- Alpha
- Beta
- Maximum Drawdown

Purpose:

- Compare funds side by side
- Support performance evaluation
- Enable ranking and filtering

NAV vs Benchmark Analysis

A line chart was developed to compare:

- Fund NAV
- Benchmark Index Performance

Purpose:

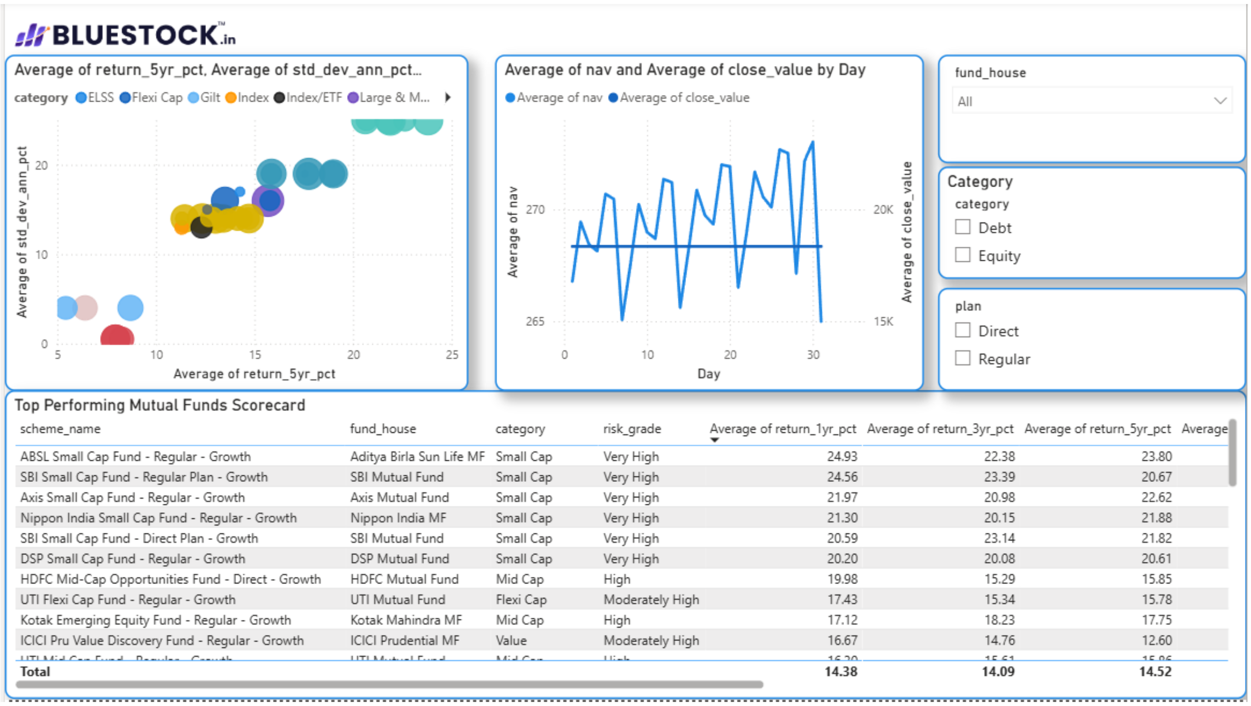
- Evaluate benchmark outperformance
- Visualize long-term growth trends

Slicers

Interactive slicers were added for:

- Fund House
- Category
- Plan Type

These filters allow users to perform customized analysis.



13.4 Investor Analytics Page

The Investor Analytics page focuses on investor participation, transaction activity, and demographic behavior.

Transaction Amount by State

A bar chart was used to compare investment activity across different states.

Purpose:

- Analyze regional participation
- Identify high-investment regions

Transaction Type Distribution

A donut chart was used to visualize:

- SIP Investments
- Lumpsum Investments
- Redemptions

Purpose:

- Understand transaction composition
- Compare investment behavior

Average SIP Amount by Age Group

A bar chart was used to compare investment activity across age groups.

Purpose:

- Identify dominant investor segments
- Analyze demographic trends

Transaction Volume Trend

A line chart was used to track transaction activity over time.

Purpose:

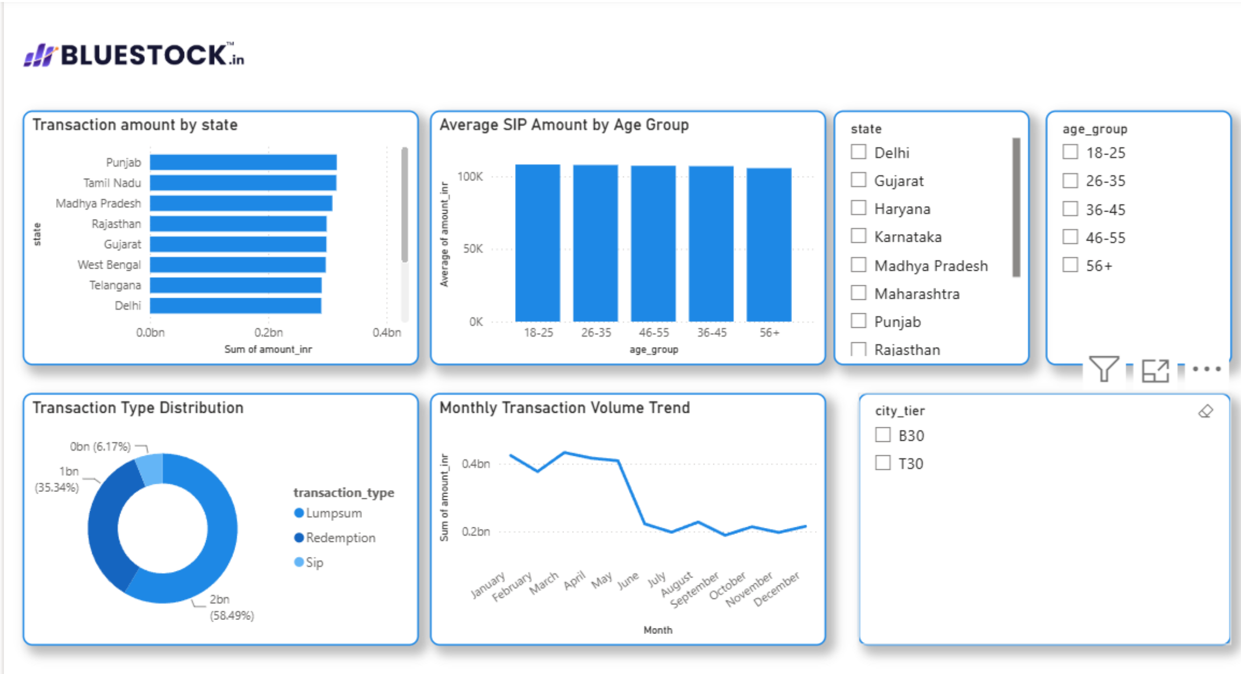
- Monitor investment trends
- Evaluate participation growth

Slicers

Interactive filters were added for:

- State
- Age Group
- City Tier

These controls improve analytical flexibility and user experience.



13.5 SIP & Market Trends Page

The SIP & Market Trends page focuses on investment flows, market performance, and category-level analysis.

SIP Inflow vs Nifty 50

A dual-axis visualization was developed using:

- SIP Inflows (Bar Chart)
- Nifty 50 Index Performance (Line Chart)

Purpose:

- Compare investor participation with market movements
- Evaluate market sentiment

Category Inflow Heatmap

A heatmap was created using conditional formatting.

Purpose:

- Compare category-level inflows
- Identify investment preferences

- Visualize fund flow intensity

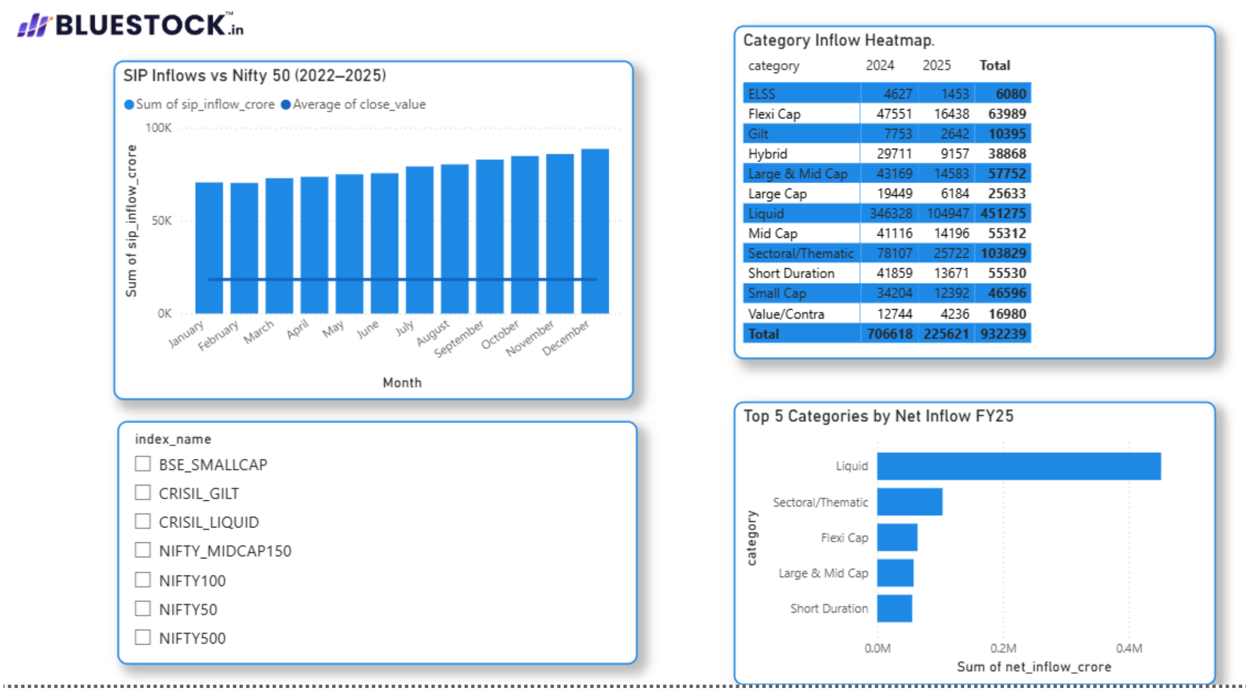
Top Categories by Net Inflow

A ranked bar chart was used to display:

- Top 5 Categories by Net Inflow

Purpose:

- Identify leading investment categories
- Analyze investor allocation patterns



13.6 Interactivity Features

Several interactive capabilities were implemented to improve user experience and analytical depth.

Drill-Through Navigation

Users can drill from the Fund Performance page into a dedicated NAV Detail page.

Purpose:

- Scheme-level analysis
- Detailed NAV trend evaluation

Dynamic Slicers

Interactive filters enable customized analysis based on:

- Fund House
- Category
- Plan Type
- State
- Age Group
- City Tier

Tooltips

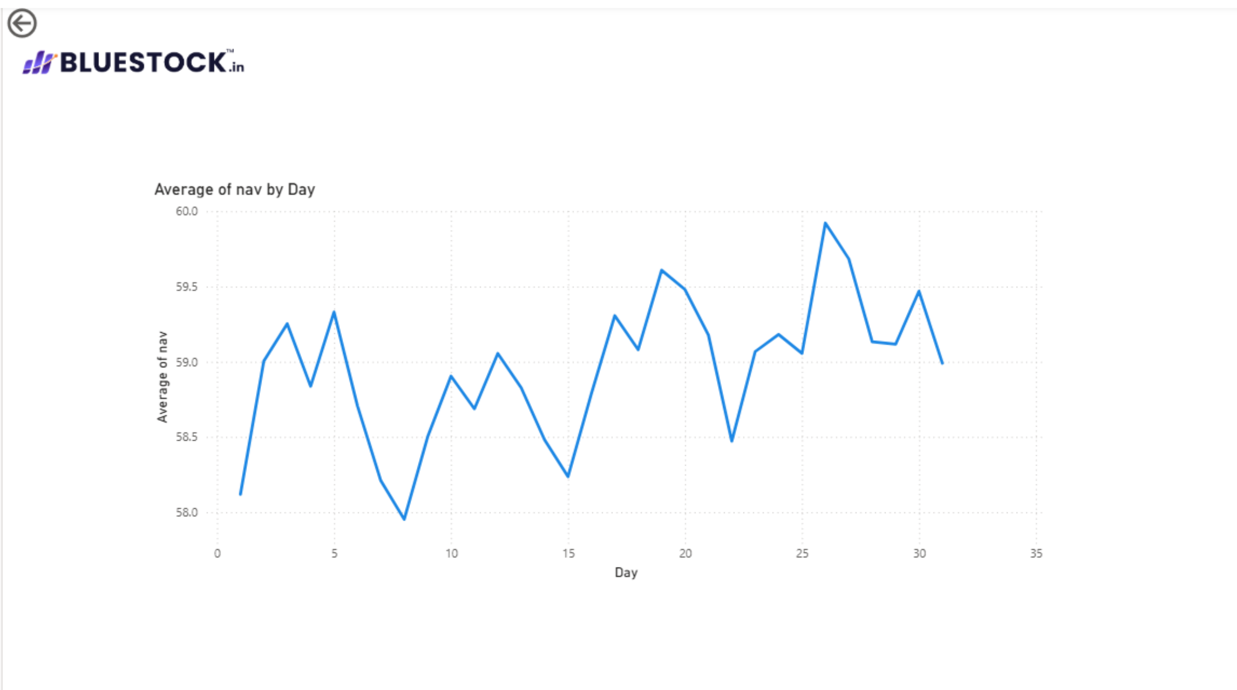
Custom tooltips were added to visualizations.

Purpose:

- Display additional contextual information
- Improve data interpretation

Cross-Filtering

Visuals interact dynamically with one another, allowing users to explore relationships between different datasets.



13.7 Dashboard Design Principles

Several design principles were followed during dashboard development.

Simplicity

Visuals were designed to communicate information clearly without unnecessary complexity.

Consistency

Uniform formatting, colors, fonts, and layouts were maintained throughout the dashboard.

Interactivity

Users can perform self-service analysis through filters, drill-through functionality, and cross-highlighting.

Business Focus

Visualizations were selected based on their ability to support decision-making and insight generation.

Branding

A Bluestock-inspired color theme and professional layout were applied to enhance visual appeal and presentation quality.

13.8 Key Dashboard Outcomes

The Power BI dashboard successfully transformed analytical findings into an interactive decision-support system.

Major outcomes include:

- Centralized visualization of mutual fund industry data.
- Efficient monitoring of AUM, SIPs, and folio growth.
- Comprehensive evaluation of fund performance.
- Improved understanding of investor behavior.
- Category-level and market trend analysis.
- Interactive exploration through slicers and drill-through pages.
- Enhanced accessibility of business insights through visual analytics.

The dashboard serves as the primary user interface of the Mutual Fund Analytics Platform and demonstrates the practical application of business intelligence techniques in the financial services domain.

14. Key Findings & Insights

14.1 Industry Growth Insights

The analysis revealed strong and consistent growth within the Indian mutual fund industry. Assets Under Management (AUM) demonstrated a significant upward trend over the analysis period, indicating increasing investor participation and confidence in mutual fund investments.

Key observations include:

- Industry AUM showed sustained growth across reporting periods.
- SIP contributions emerged as a major driver of industry expansion.
- Investor folio counts increased steadily, reflecting broader market participation.
- Growth in mutual fund adoption highlighted increasing awareness of financial planning and long-term investing.

These findings indicate a healthy and expanding mutual fund ecosystem supported by both retail and institutional participation.

14.2 Fund Performance Insights

Performance analytics revealed notable differences in fund returns, risk characteristics, and benchmark outperformance.

Key observations include:

- Several schemes demonstrated strong long-term return performance.
- Risk-adjusted metrics such as Sharpe Ratio and Sortino Ratio provided more meaningful comparisons than absolute returns alone.
- Funds with higher Sharpe Ratios generally delivered better returns relative to the level of risk undertaken.
- Alpha analysis identified schemes that consistently outperformed benchmark expectations.
- Maximum Drawdown analysis highlighted significant variations in downside risk exposure across funds.

- Benchmark comparison revealed that only a subset of schemes consistently generated excess returns.

These insights emphasize the importance of evaluating both returns and risk when assessing mutual fund performance.

14.3 Investor Behavior Insights

Investor transaction analysis provided valuable insights into participation patterns and investment behavior.

Key observations include:

- Investment activity varied significantly across states and regions.
- Certain states contributed a larger share of transaction volume.
- SIP investments represented a substantial portion of overall investor activity.
- Different age groups exhibited varying investment preferences and participation levels.
- Urban investors demonstrated higher participation rates compared to smaller city tiers.
- Long-term investment behavior was increasingly driven by systematic investment plans.

These findings highlight the growing diversity of the investor base and the increasing adoption of disciplined investment practices.

14.4 SIP & Market Insights

The relationship between SIP activity and market performance was analyzed using SIP inflow and benchmark datasets.

Key observations include:

- SIP inflows remained relatively stable despite fluctuations in market performance.
- Investor participation continued to grow during both bullish and volatile market conditions.
- Category-level inflows revealed changing investor preferences across fund segments.
- Certain categories consistently attracted higher inflows than others.
- Market benchmarks exhibited long-term growth despite short-term corrections.

- Investors increasingly favored systematic investing approaches rather than attempting to time the market.

The findings demonstrate the resilience of SIP-based investing and its role in supporting long-term wealth creation.

14.5 Strategic Recommendations

Based on the analytical findings, the following recommendations are proposed:

For Investors

- Evaluate funds using both return and risk-adjusted metrics.
- Focus on long-term investment horizons rather than short-term market movements.
- Consider diversification across multiple fund categories.
- Utilize SIPs as a disciplined investment strategy.

For Fund Management Companies

- Improve investor education initiatives.
- Expand outreach programs in underrepresented regions.
- Promote category diversification to reduce concentration risk.
- Enhance transparency in fund performance communication.

For Future Analytics Initiatives

- Integrate real-time market and NAV data.
- Implement predictive analytics for performance forecasting.
- Develop personalized fund recommendation systems.
- Introduce automated reporting and dashboard refresh mechanisms.

These recommendations can help improve investment decision-making, strengthen investor participation, and support continued growth within the mutual fund industry.

15. Challenges Faced

15.1 Data Collection Challenges

The project involved integrating multiple datasets covering various aspects of the mutual fund industry, including fund information, NAV history, performance metrics, investor transactions, SIP statistics, benchmark indices, AUM data, and folio records.

One of the primary challenges during data collection was ensuring that all datasets were available in a consistent and usable format. Since the datasets originated from different domains, variations in structure, naming conventions, and data granularity were observed.

Key challenges included:

- Managing multiple datasets simultaneously.
- Understanding relationships between datasets.
- Ensuring dataset completeness before analysis.
- Maintaining consistency across different data sources.

These challenges required careful planning before proceeding to data preparation and integration stages.

15.2 Data Quality Challenges

Data quality issues represented one of the most significant challenges during the project.

Several datasets contained:

- Missing values
- Duplicate records
- Inconsistent formatting
- Invalid data types
- Incomplete observations

Date fields required additional attention because different datasets followed different date formats.

To address these issues, extensive data cleaning and validation procedures were implemented using Python and Pandas.

Data quality improvements included:

- Missing value treatment
- Duplicate removal
- Data type correction
- Format standardization
- Validation checks

These activities significantly improved dataset reliability and analytical accuracy.

15.3 Data Integration Challenges

Integrating multiple datasets into a unified analytical environment presented several challenges.

Each dataset represented a different aspect of the mutual fund ecosystem, making relationship mapping an important task.

Challenges included:

- Establishing relationships between datasets.
- Identifying common keys for integration.
- Maintaining referential consistency.
- Aligning time-series data across multiple sources.
- Creating a unified analytical model.

Special attention was required when linking fund-level, transaction-level, and industry-level datasets.

The implementation of a structured database model and Power BI relationships helped overcome these challenges.

15.4 Dashboard Development Challenges

Developing an interactive Power BI dashboard involved both technical and design-related challenges.

Initially, creating relationships between datasets and ensuring accurate filtering behavior required multiple iterations and validation checks.

Additional challenges included:

- Designing effective dashboard layouts.
- Selecting appropriate visualizations.
- Implementing cross-filtering functionality.
- Configuring drill-through navigation.
- Managing dashboard performance.
- Ensuring consistency across pages.

Several visualizations required customization to accurately represent analytical findings while maintaining readability and user experience.

Through continuous testing and refinement, an interactive and business-focused dashboard was successfully developed.

15.5 Key Learnings

The project provided valuable practical experience across multiple areas of data analytics and business intelligence.

Major learnings include:

Data Engineering

- Building ETL pipelines.
- Data cleaning and preprocessing.
- Dataset integration techniques.

Database Management

- Designing relational databases.
- Writing analytical SQL queries.
- Managing structured data efficiently.

Data Analytics

- Performing exploratory data analysis.
- Calculating financial performance metrics.
- Applying advanced analytical techniques.

Business Intelligence

- Developing interactive Power BI dashboards.
- Creating KPI-driven visualizations.
- Implementing drill-through and filtering features.

Problem Solving

- Handling real-world data quality issues.
- Resolving integration challenges.

- Converting analytical findings into actionable insights.

Overall, the project provided hands-on experience in applying data engineering, analytics, SQL, and business intelligence concepts to solve a real-world financial analytics problem. The knowledge gained throughout the project significantly strengthened both technical and analytical skills.

Bonus Challenges

B1 – Automated ETL Scheduling

An automated ETL pipeline was implemented to fetch the latest Net Asset Value (NAV) data from mfapi.in every weekday at 8:00 PM. The automation was configured using Windows Task Scheduler. The ETL process performs data extraction, transformation, validation, and loading into the SQLite database without manual intervention. This ensures that the analytical database remains continuously updated with the latest mutual fund information.

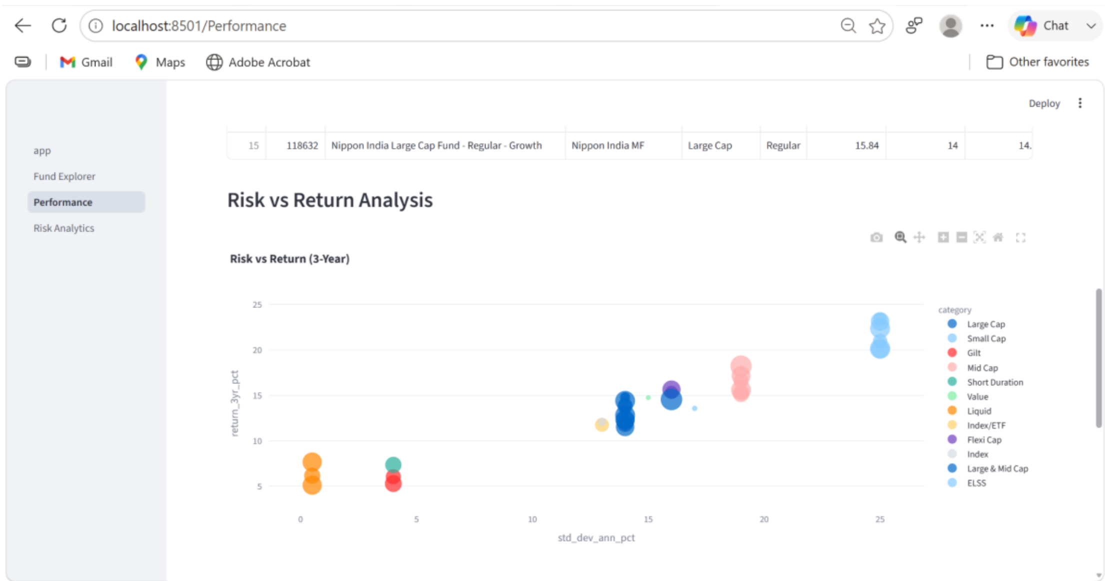
OUTPUT:

```
date,nav,fetch_date
11-06-2026,105.57370,2026-06-12 17:35:18.584285
10-06-2026,105.65690,2026-06-12 17:35:18.584285
09-06-2026,105.57930,2026-06-12 17:35:18.584285
08-06-2026,105.30840,2026-06-12 17:35:18.584285
05-06-2026,105.10700,2026-06-12 17:35:18.584285
04-06-2026,104.80950,2026-06-12 17:35:18.584285
03-06-2026,104.74960,2026-06-12 17:35:18.584285
02-06-2026,104.76630,2026-06-12 17:35:18.584285
01-06-2026,104.70250,2026-06-12 17:35:18.584285
29-05-2026,104.65700,2026-06-12 17:35:18.584285
27-05-2026,104.58330,2026-06-12 17:35:18.584285
26-05-2026,104.52690,2026-06-12 17:35:18.584285
```

B2 – Streamlit Web Application

A Streamlit-based web application was developed as an alternative to the Power BI dashboard. The application provides an interactive interface for exploring mutual fund data, performance metrics, risk analytics, and fund recommendations. Users can filter funds, view performance visualizations, analyze risk measures, and download recommendation reports. The application demonstrates real-time analytics capabilities using Python and SQLite.

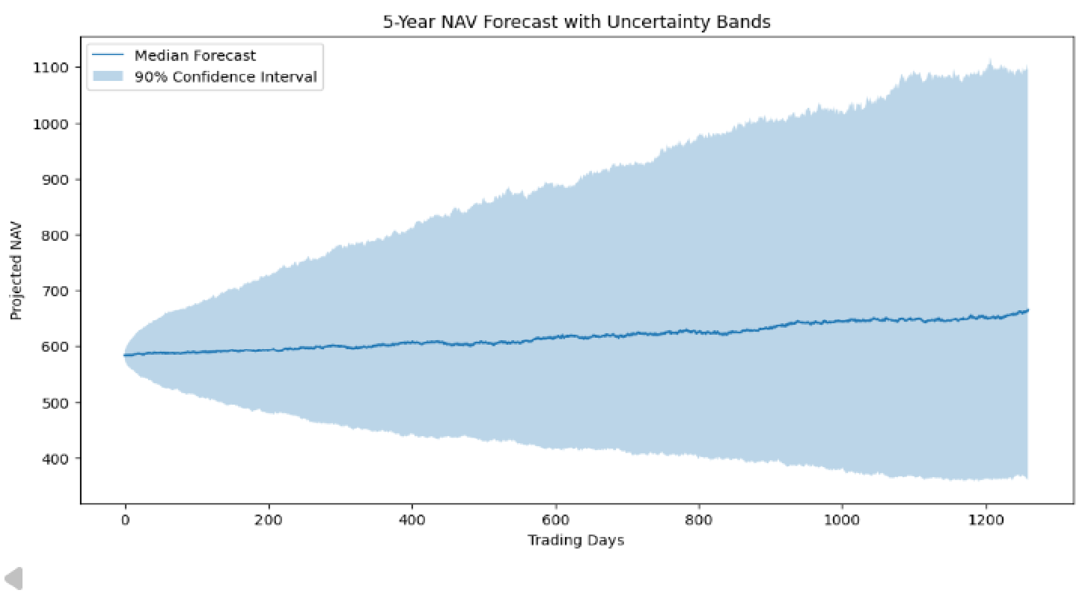
OUTPUT:



B3 – Monte Carlo Simulation

A Monte Carlo simulation model was developed to project mutual fund NAV growth over a five-year horizon. Historical NAV returns were used to estimate expected return and volatility. A total of 1,000 simulation paths were generated to model future uncertainty. Confidence intervals were calculated and visualized to illustrate the range of potential future outcomes and associated investment risk.

OUTPUT:

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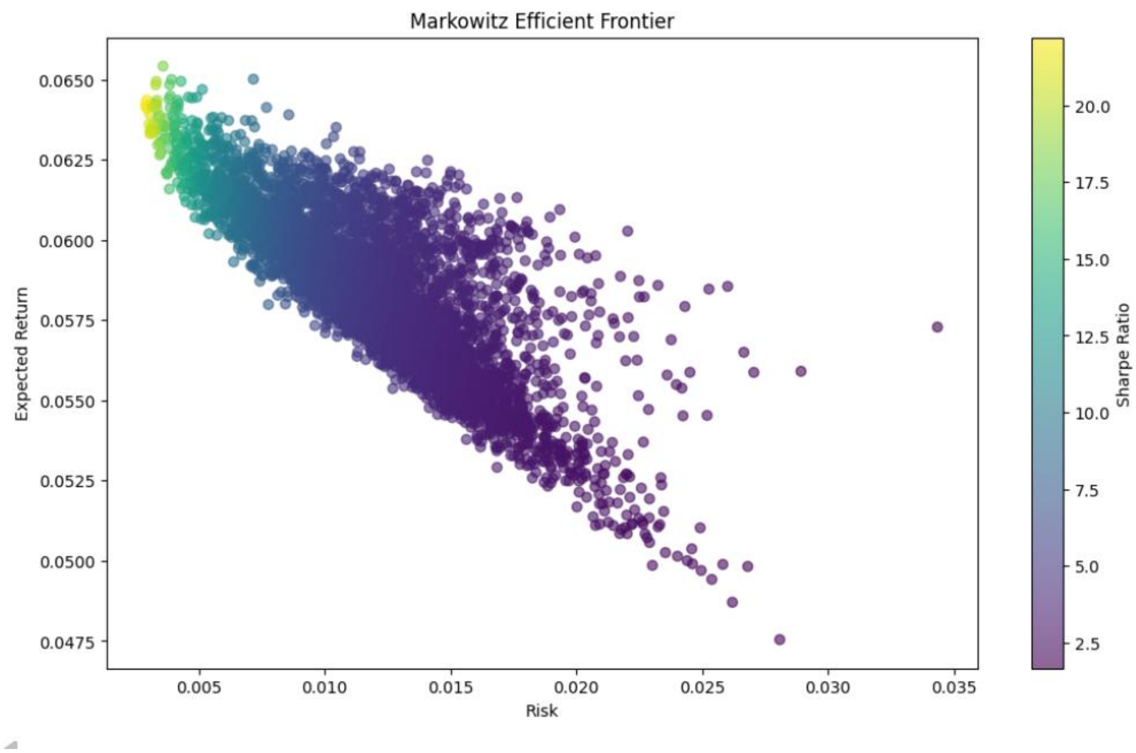
&lt;Figure size 640x480 with 0 Axes&gt;
```

B4 – Markowitz Efficient Frontier Portfolio Optimization

Portfolio optimization was performed using Modern Portfolio Theory. Five selected mutual funds were used to generate 5,000 random portfolio combinations. Expected returns, portfolio risk, and Sharpe ratios were calculated for each portfolio. The Efficient Frontier was visualized, and the portfolio with the maximum Sharpe ratio was identified as the optimal allocation. This analysis demonstrates risk-return optimization techniques used in investment management.



B5 – Automated HTML Email Reporting

An automated HTML email reporting system was developed to generate weekly mutual fund performance summaries. The solution extracts performance metrics from the SQLite database, creates formatted HTML reports, and distributes them via email using SMTP. The reporting process can be scheduled automatically using Windows Task Scheduler, enabling periodic performance monitoring and reporting.

OUTPUT:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```
PS D:\bluestock_mf_capstone> python D:\bluestock_mf_capstone\BONUS_CHALLENGES\B5\weekly_email_repo
rt.py
Email Sent Successfully
PS D:\bluestock_mf_capstone>
```

Weekly Mutual Fund Performance Report

srinivasan2004ss@gmail.com
To: You

Fri 2026-06-12 17:44

Weekly Mutual Fund Performance Report

Top 10 Funds Based on 3-Year Returns

scheme_name	return_3yr_pct	sharpe_ratio	aum_crore
SBI Small Cap Fund - Regular Plan - Growth	23.39	0.94	19259
SBI Small Cap Fund - Direct Plan - Growth	23.14	0.93	36061
ABSL Small Cap Fund - Regular - Growth	22.38	0.90	41613
Axis Small Cap Fund - Regular - Growth	20.98	0.84	21545
Nippon India Small Cap Fund - Regular - Growth	20.15	0.81	43630
DSP Small Cap Fund - Regular - Growth	20.08	0.80	35124
Kotak Emerging Equity Fund - Regular - Growth	18.23	0.96	47469
ICICI Pru Midcap Fund - Regular - Growth	18.08	0.95	979
DSP Midcap Fund - Regular - Growth	17.16	0.90	37835
HDFC Mid-Cap Opportunities Fund - Regular - Growth	16.58	0.87	23185